



SELF-TESTING IN QUANTUM INFORMATION THEORY

TUTORIAL



IVAN ŠUPIĆ

CNRS, LIG

CEQIP, MIKULOV, 14.09.2026



ALICE



BOB



ALICE



BOB



EVE

STRUCTURE

- ACT I : THE CONCEPT OF SELF-TESTING
- ACT II : MAIN TECHNIQUES
- ACT III : SOME APPLICATIONS

★ACT 1 : THE CONCEPT OF SELF-TESTING

TERMINOLOGY : SELF-TESTING

JOURNAL OF COMPUTER AND SYSTEM SCIENCES **47**, 549–595 (1993)

Self-Testing/Correcting with Applications to Numerical Problems

MANUEL BLUM*

Computer Science Division, U.C. Berkeley, Berkeley, California 94720

MICHAEL LUBY†

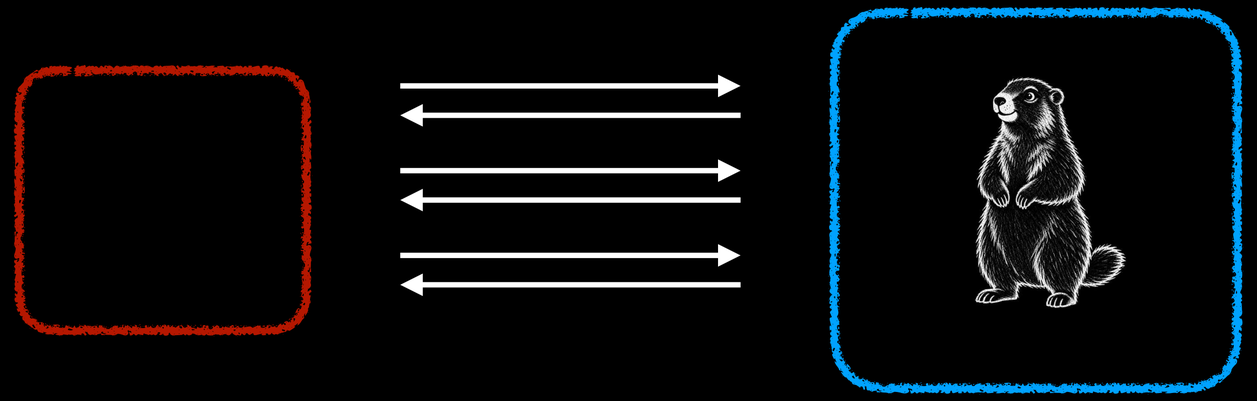
International Computer Science Institute, Berkeley, California 94704; and Computer Science Division, U.C. Berkeley, Berkeley, California 94720

AND

RONITT RUBINFELD‡

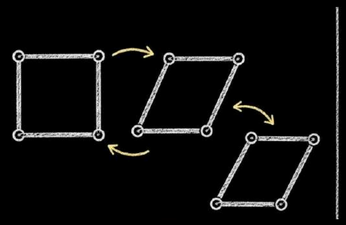
Department of Computer Science, Cornell University, Ithaca, New York 14853

Received August 14, 1990; revised November 19, 1991

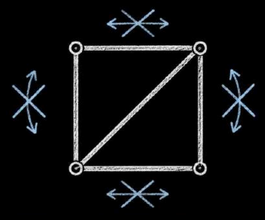


TERMINOLOGY : RIGIDITY

FLEXIBLE



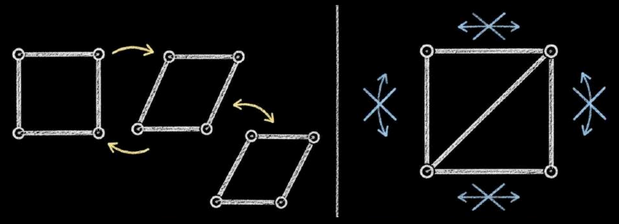
RIGID



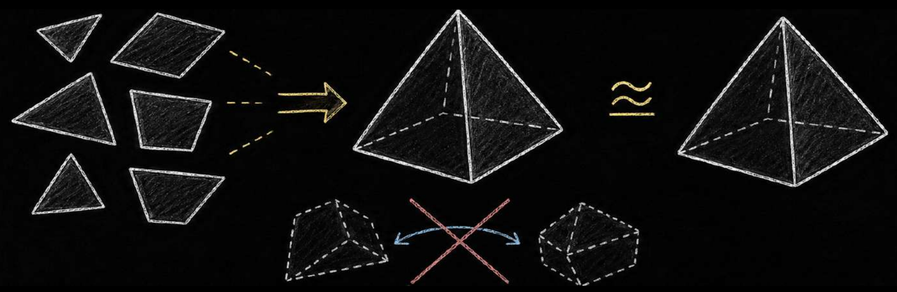
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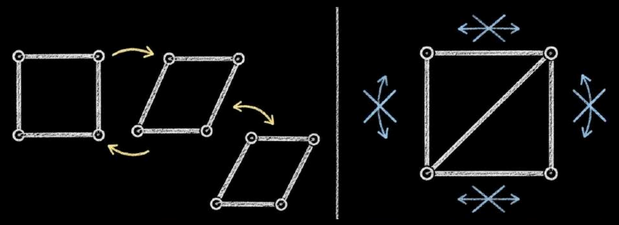
CAUCHY RIGIDITY



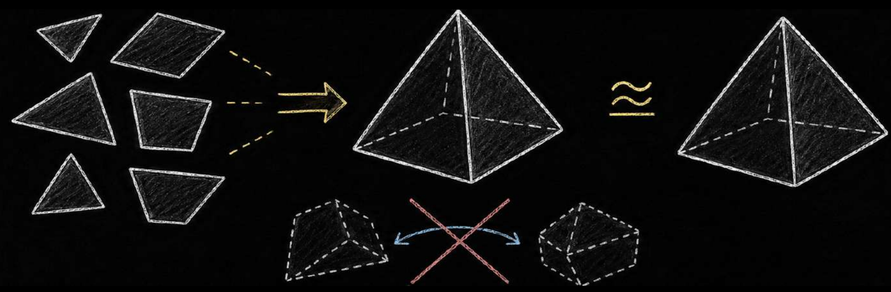
A CONVEX POLYHEDRON IS DETERMINED BY ITS FACES, UP TO CONGRUENCE

TERMINOLOGY : RIGIDITY

FLEXIBLE | RIGID

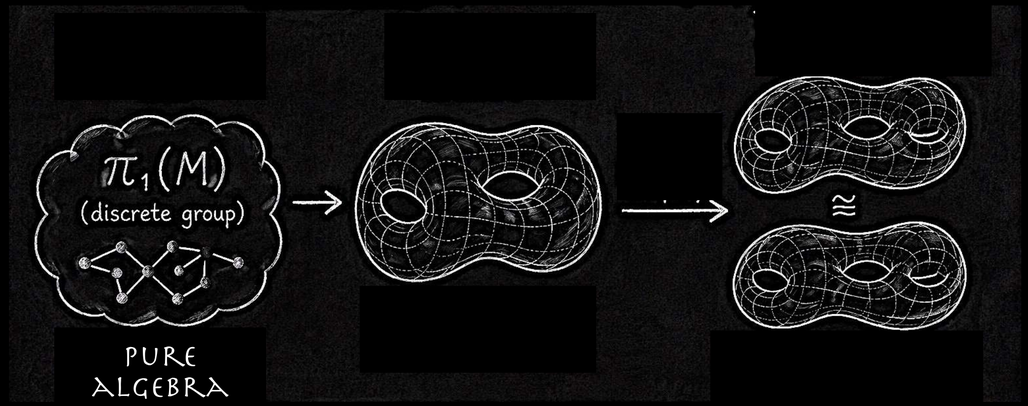


CAUCHY RIGIDITY



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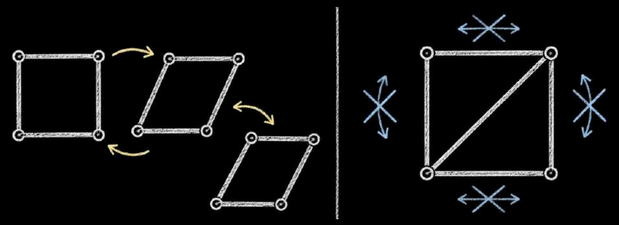
MOSTOW RIGIDITY



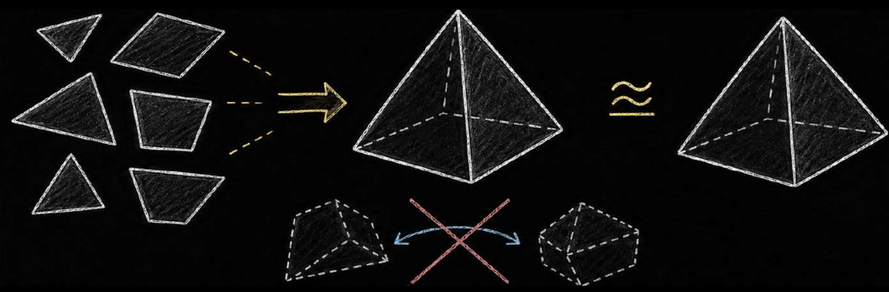
A CLOSED HYPERBOLIC MANIFOLD OF DIMENSION LARGER THAN 3 IS DETERMINED UP TO ISOMETRY BY ITS FUNDAMENTAL GROUP

TERMINOLOGY : RIGIDITY

FLEXIBLE | RIGID

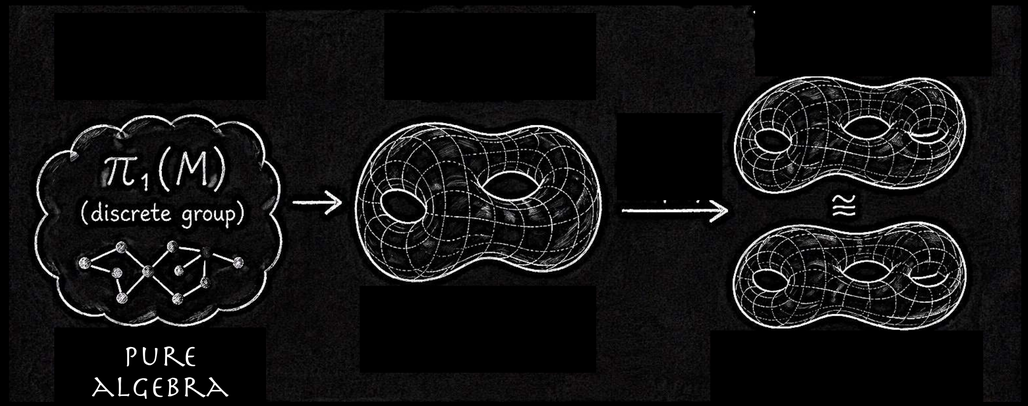


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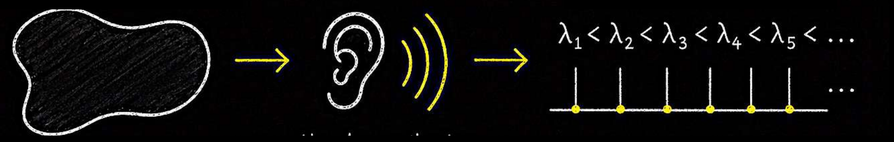
MOSTOW RIGIDITY



A CLOSED HYPERBOLIC MANIFOLD OF DIMENSION LARGER THAN 3 IS DETERMINES UP TO ISOMETRY BY ITS FUNDAMENTAL GROUP

SPECTRAL RIGIDITY

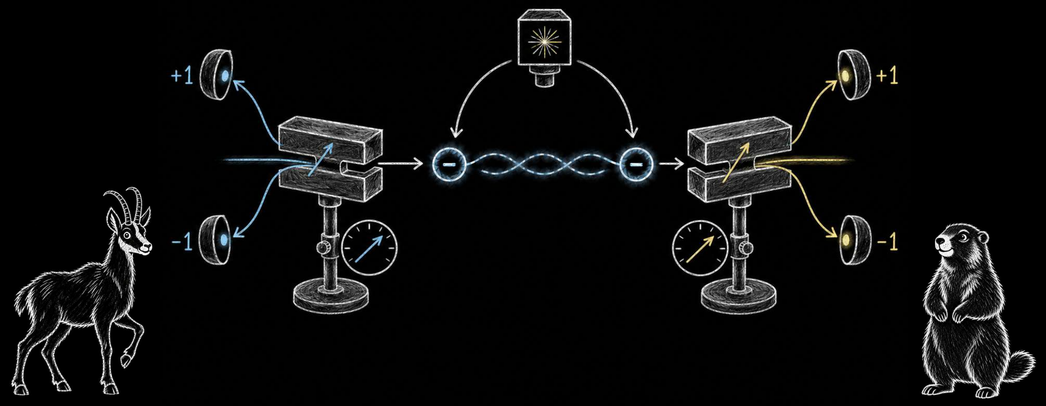
KAC: CAN ONE HEAR THE SHAPE OF A DRUM?



DRUM SHAPE

LAPLACIAN SPECTRUM

CAN WE HEAR THE SOUND OF QUANTUM STATES FROM CORRELATIONS?

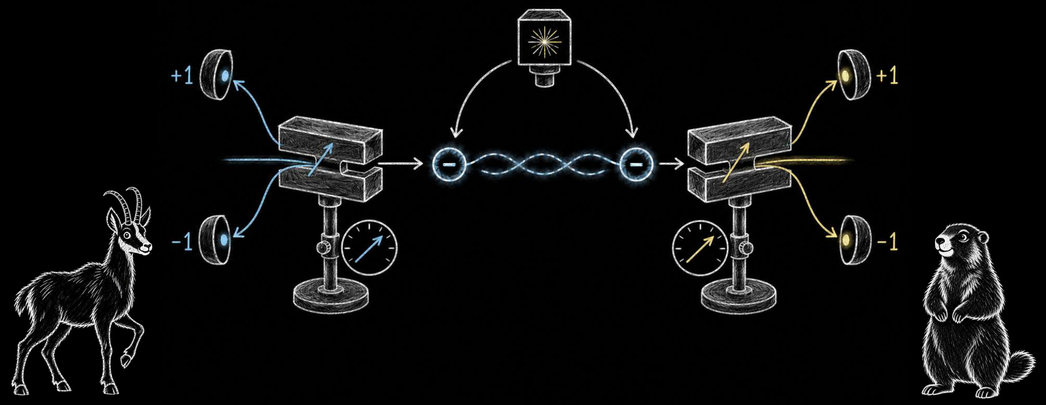


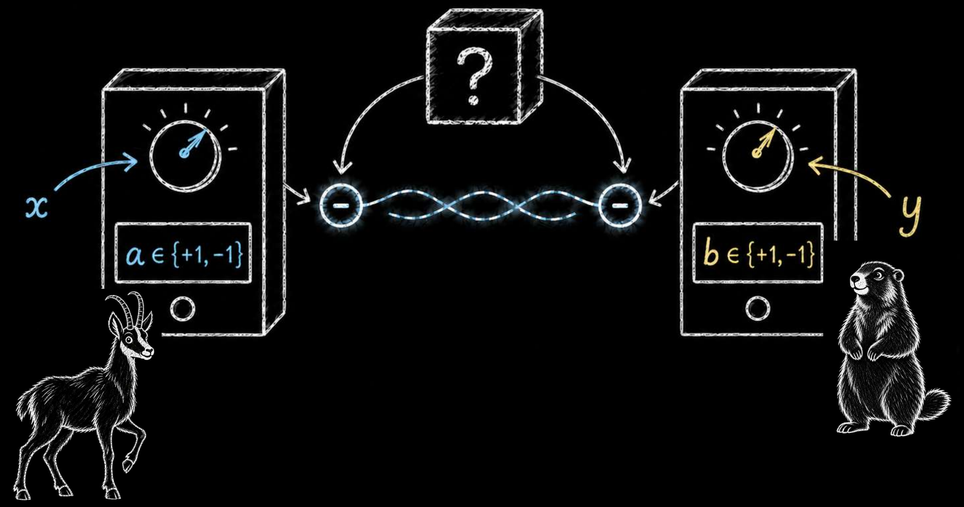
$$|\mathrm{EPR}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$A_0 = \sigma_z \qquad B_0 = \frac{\sigma_z + \sigma_x}{\sqrt{2}}$$

$$A_1 = \sigma_x \qquad B_1 = \frac{\sigma_z - \sigma_x}{\sqrt{2}}$$

CAN WE HEAR THE SOUND OF QUANTUM STATES FROM CORRELATIONS?



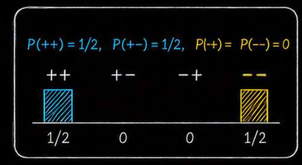


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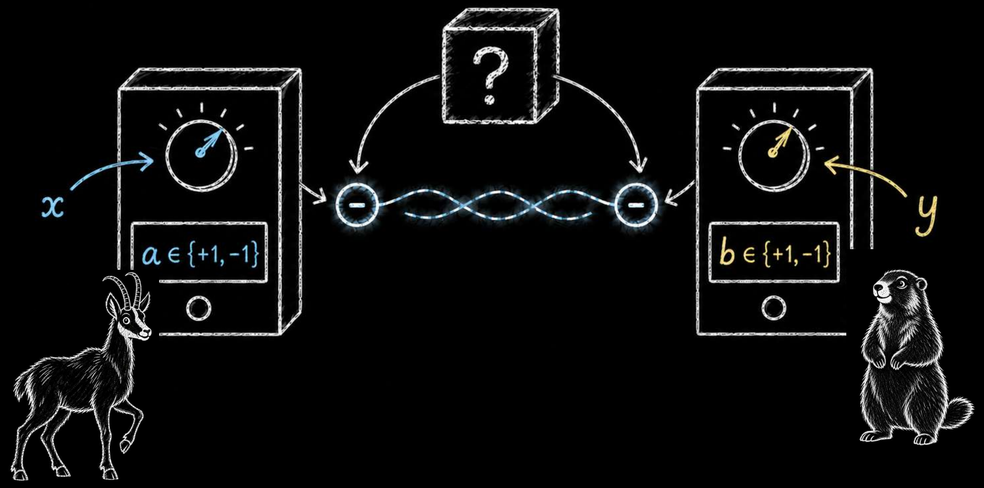
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$$p(a, b \,|\, x, y)$$



NONLOCAL GAMES / BELL INEQUALITIES



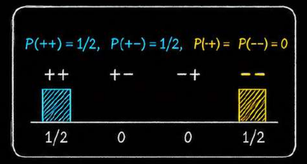
$$\omega = \sum_{x,y} \pi(x,y) \sum_{a,b} V(a,b \mid x,y)\, p(a,b \mid x,y)$$

$$\omega = \sum_{x,y,a,b} \pi(x,y)\, V(a,b \mid x,y)\, \langle \psi | M_a^x \otimes N_b^y | \psi \rangle \, .$$

ω_c CLASSICAL VALUE

ω_q QUANTUM VALUE

$$p(a,b \mid x,y)$$



BELL VALUE

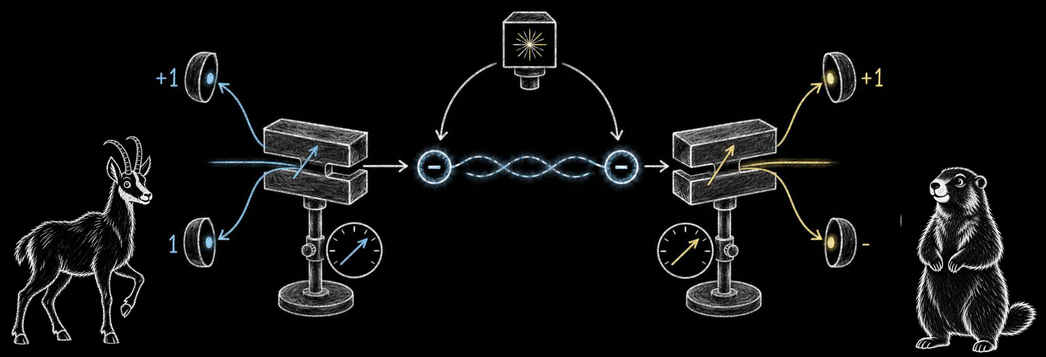
$$\beta = \sum_{x,y,a,b} b_{a,b|x,y}\, p(a,b \mid x,y)$$

BELL OPERATOR

$$\mathcal{B} = \sum_{x,y,a,b} b_{a,b|x,y}\, M_a^x \otimes N_b^y$$

$$\beta = \langle \psi | \mathcal{B} | \psi \rangle$$

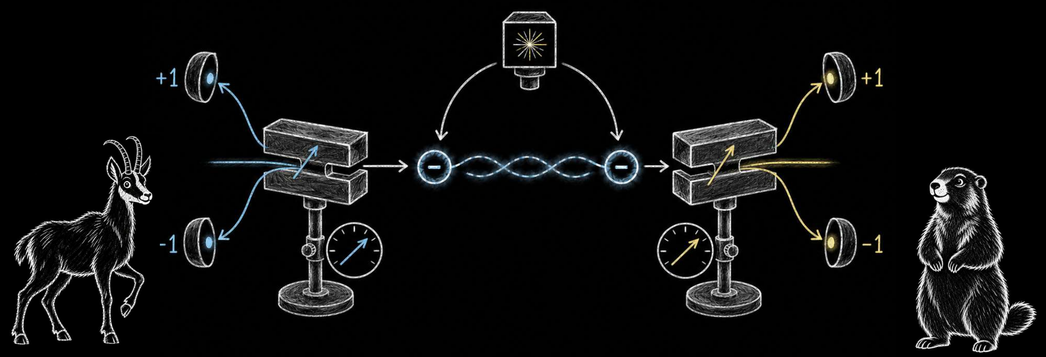
CAN WE HEAR THE SOUND OF QUANTUM STATES FROM CORRELATIONS?



CAN WE TRUST OUR SENSES?

WHAT ARE THESE MEASUREMENTS REALLY DOING?

CAN WE HEAR THE SOUND OF QUANTUM STATES FROM CORRELATIONS?



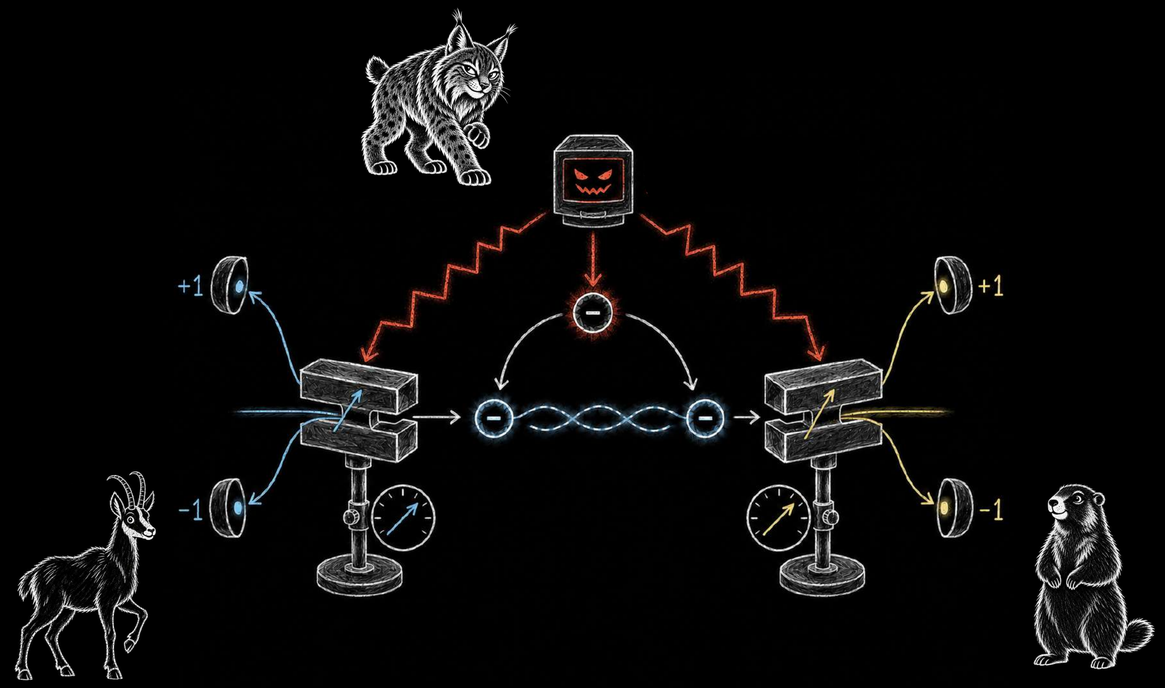
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WHAT ARE THESE MEASUREMENTS REALLY DOING?

CAN QUANTUM THEORY PASS THE TEST OF CARTESIAN METHODOLOGICAL DOUBT?

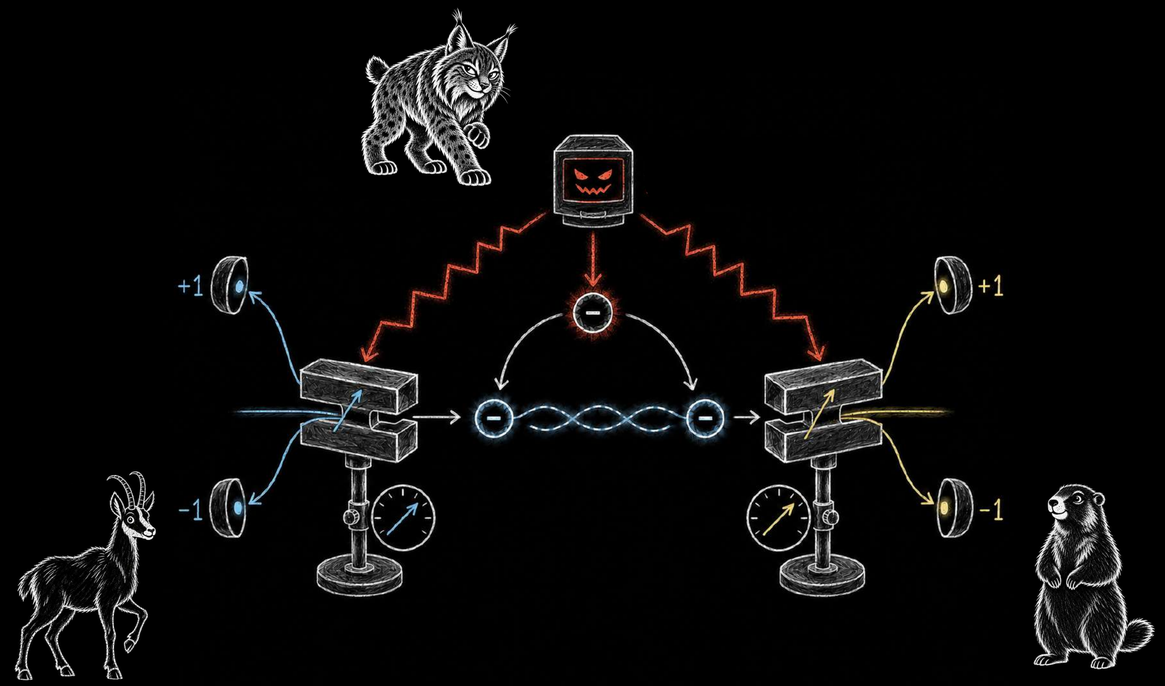


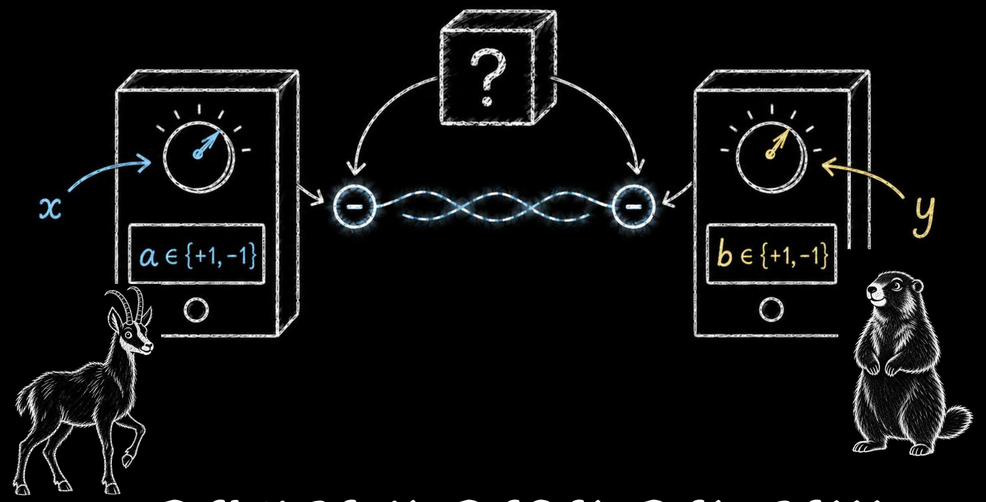
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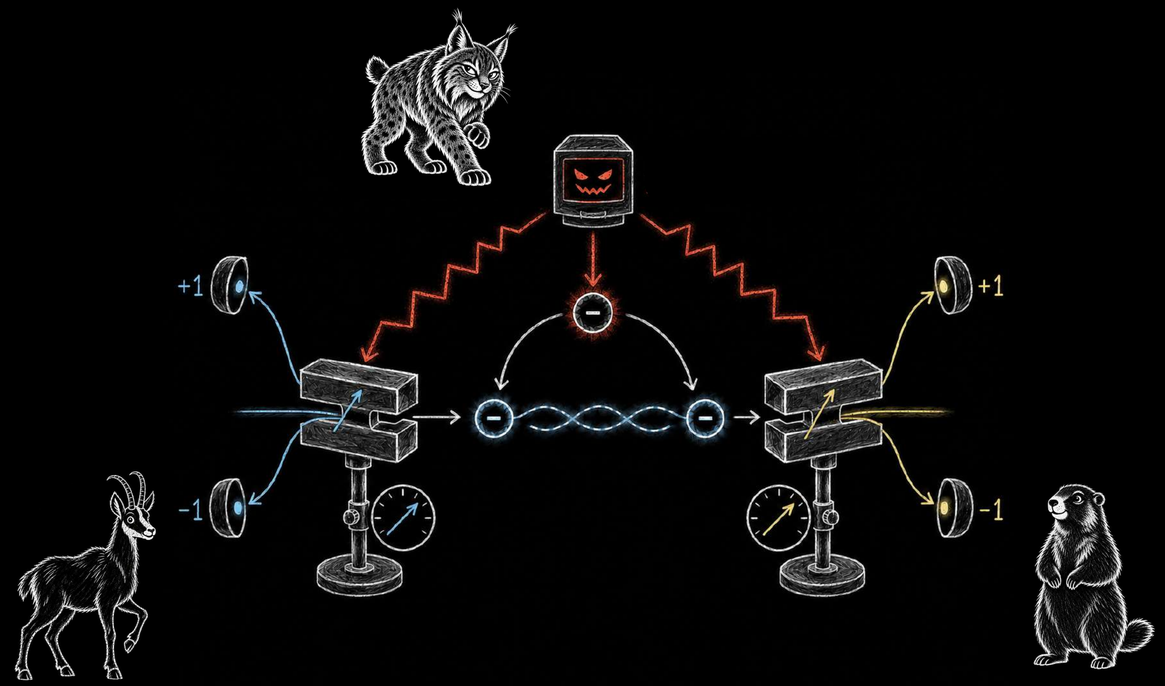


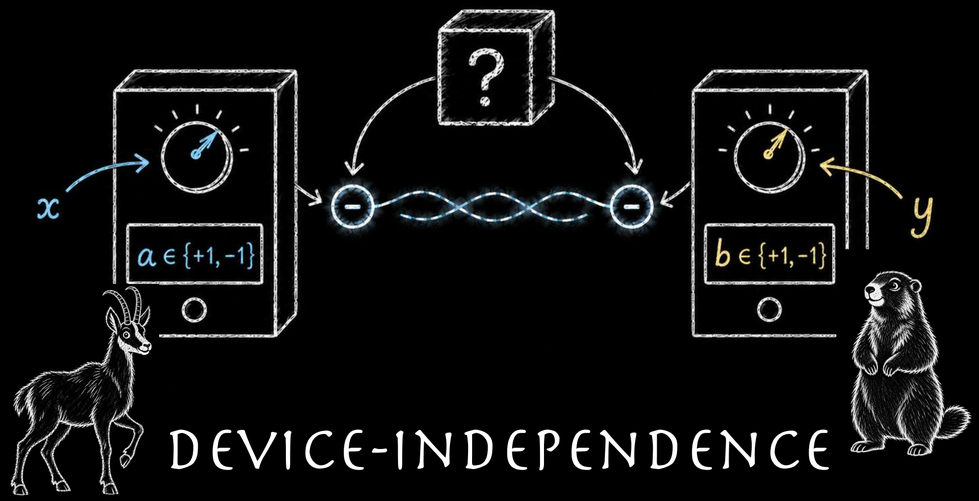
DEVICE-INDEPENDENCE!!!



Mayers, Yao
FOCS [1998]

CAN WE HEAR THE SOUND OF QUANTUM STATES FROM CORRELATIONS?





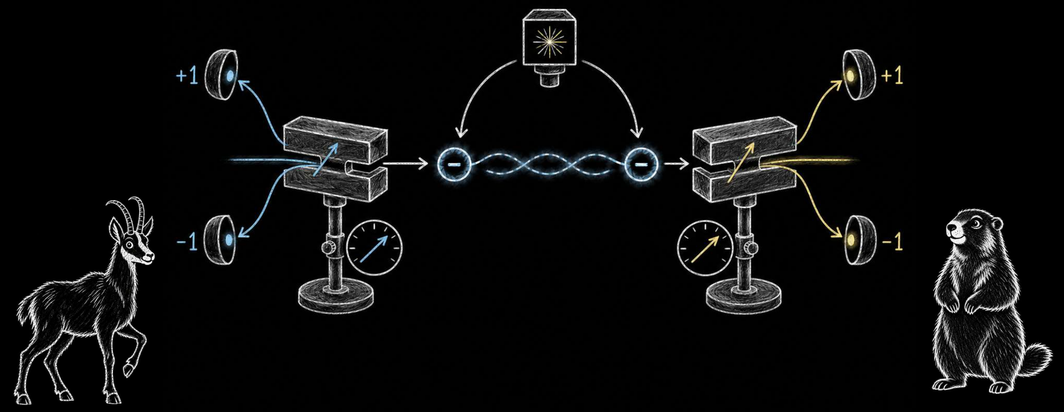
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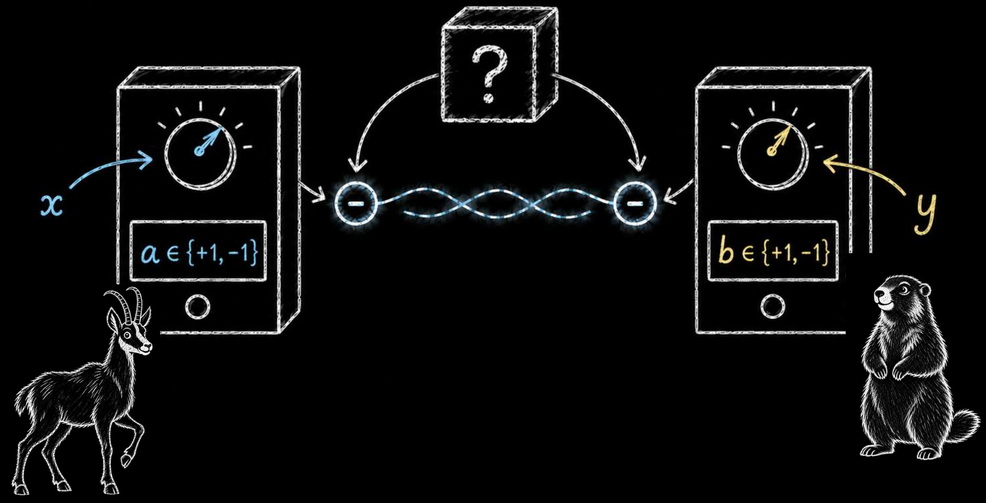
WHAT IS THERE IS AN ADVERSARY?

Mayers, Yao
FOCS [1998]

SELF-TESTING WAS NOT INVENTED TO STUDY FOUNDATIONS. IT WAS INVENTED BECAUSE SOMEONE ASKED WHETHER YOU CAN TRUST A DEVICE YOU DIDN'T BUILD.

CAN WE HEAR THE SOUND OF QUANTUM STATES FROM CORRELATIONS?





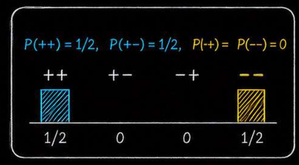
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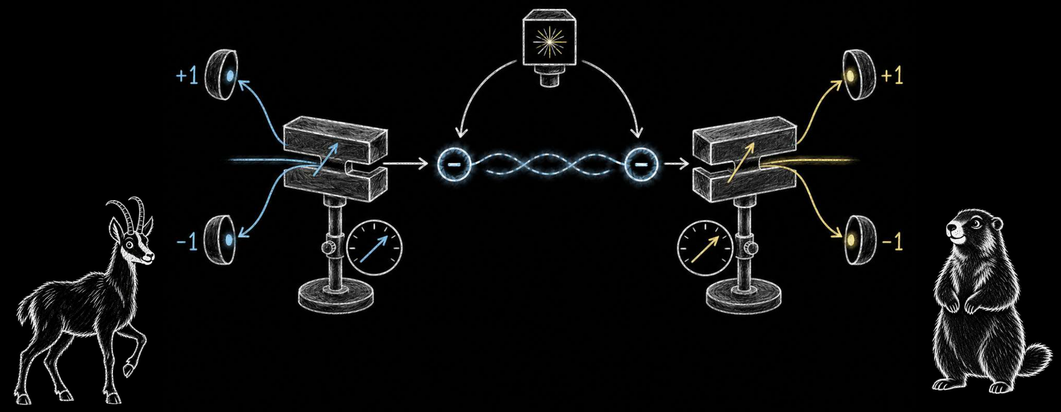
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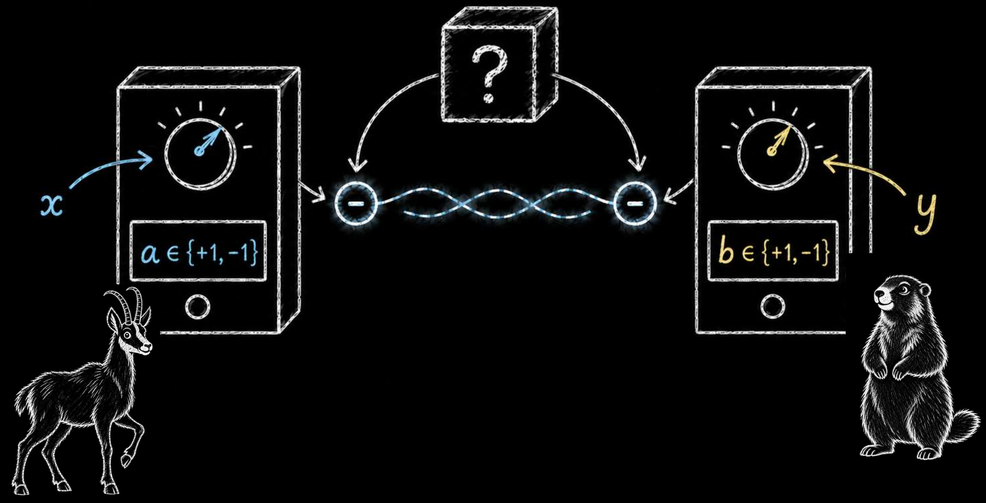
$$\langle \mathrm{EPR} | A_x \otimes B_y | \mathrm{EPR} \rangle$$

$$p(a, b \,|\, x, y)$$



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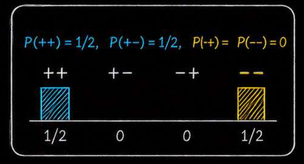
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$$\langle \mathrm{EPR} | A_x \otimes B_y | \mathrm{EPR} \rangle$$

?

$$p(a, b \,|\, x, y)$$

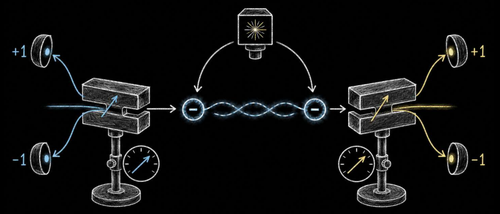


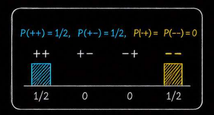
Mayers, Yao
QIC 4,4 [2004]

WHAT CAN WE HEAR?

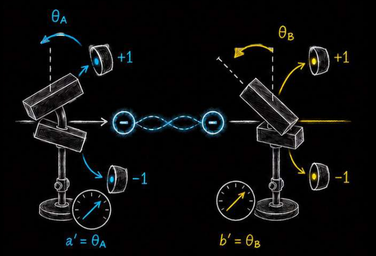
LOCAL UNITARIES

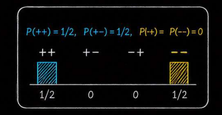
$$S = \left\{ |\psi\rangle, M_{a|x}, N_{b|y} \right\}$$





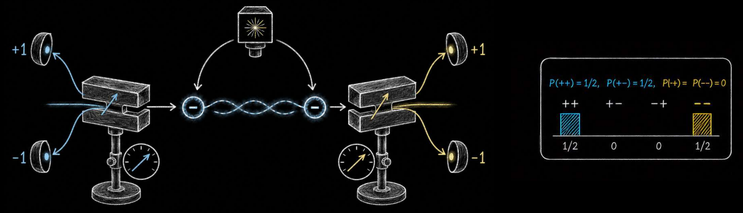
$$\tilde{S} = \left\{ U \otimes V|\psi\rangle, UM_{a|x}U^\dagger, VN_{b|y}V^\dagger \right\}$$





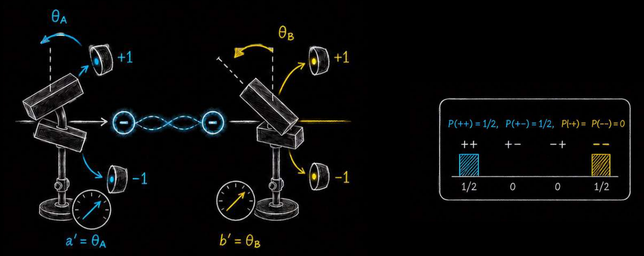
WHAT CAN WE HEAR?

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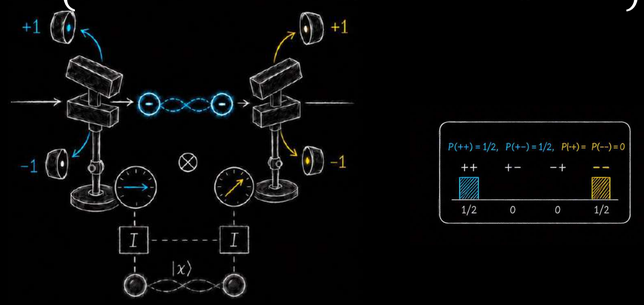
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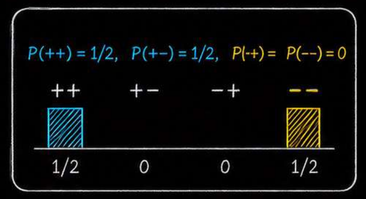
ADDITIONAL DEGREES OF FREEDOM

$$\bar{S} = \left\{ |\psi\rangle \otimes |\xi\rangle, M_{a|x} \otimes \mathbb{I}, N_{b|y} \otimes \mathbb{I} \right\}$$



WHAT CAN WE HEAR?

$p(a,b\,|\,x,y)$

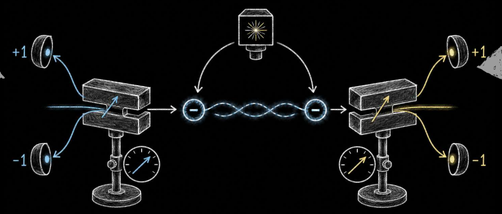


SELF-TESTING

EQUIVALENCE CLASS



LOCAL ISOMETRIES

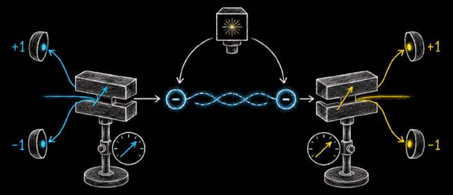


EXACT SELF-TESTING : DEFINITION

BLUEPRINT

WHAT IS ACCESSIBLE

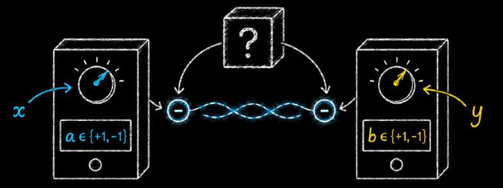
REFERENCE STRATEGY



$$S = \left\{ |\psi\rangle, M_a^x, N_b^y \right\}$$

?

PHYSICAL STRATEGY



$$\tilde{S} = \left\{ |\tilde{\psi}\rangle, \tilde{M}_a^x, \tilde{N}_b^y \right\}$$

ACCESS ONLY TO AN EVENT THE STRATEGY PRODUCES!

EVENT:
FULL BEHAVIOUR $\{p(a,b|x,y)\}$
SCORE IN A NONLOCAL GAME ω

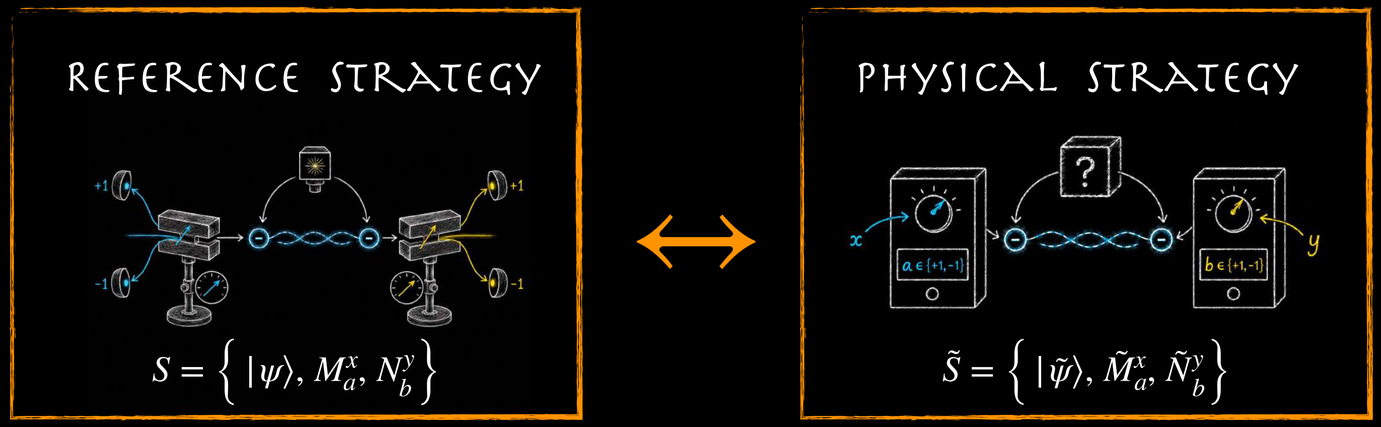
$$\Phi_A \otimes \Phi_B$$

$$\Phi_A \otimes \Phi_B \left[|\tilde{\psi}\rangle \right] = |\psi\rangle \otimes |\text{junk}\rangle$$

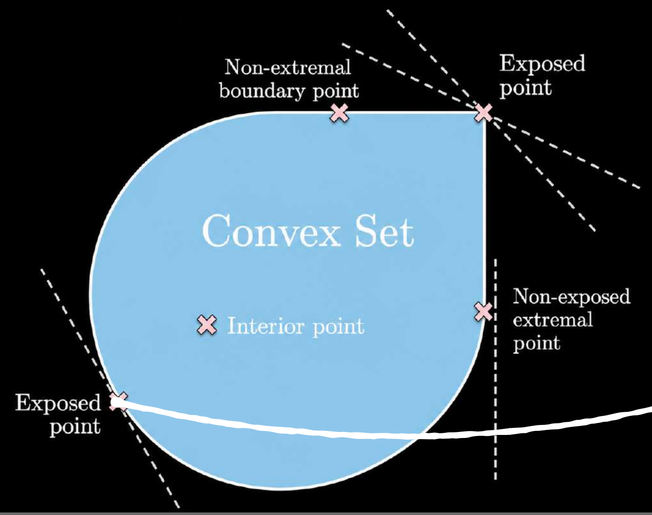
$$\Phi_A \otimes \Phi_B \left[\tilde{M}_a^x \otimes \tilde{N}_b^y |\tilde{\psi}\rangle \right] = \left(M_a^x \otimes N_b^y |\psi\rangle \right) \otimes |\text{junk}\rangle$$

AN EVENT SELF-TESTS THE REFERENCE STRATEGY S IF EVERY PHYSICAL STRATEGY $\tilde{S}$ REPRODUCING THE EVENT MUST BE RELATED TO THE REFERENCE STRATEGY S THROUGH A LOCAL ISOMETRY.

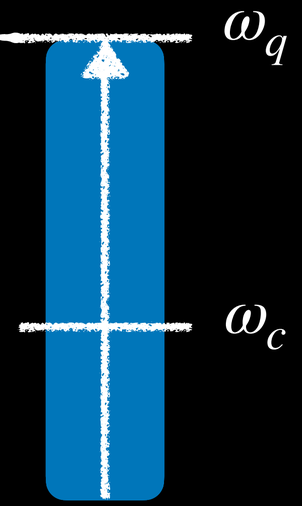
EXACT SELF-TESTING : DEFINITION



$\Phi_A \otimes \Phi_B$



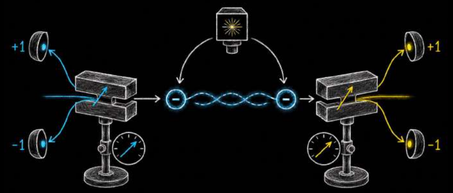
SELF-TESTING EVENT



Goh, Kaniewski, Wolfe, Vertesi, Wu, Cai, Liang, Scarani
PRA 97,022104 [2018]

ROBUST SELF-TESTING

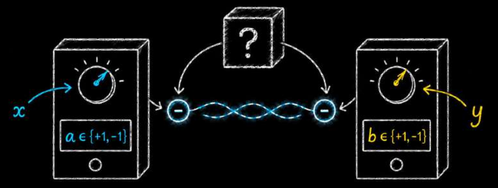
REFERENCE STRATEGY



$$S = \left\{ |\psi\rangle, M_a^x, N_b^y \right\}$$

?

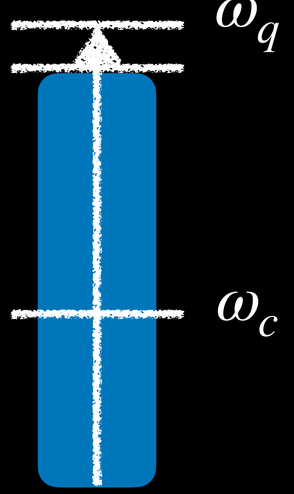
PHYSICAL STRATEGY



$$\tilde{S} = \left\{ |\tilde{\psi}\rangle, \tilde{M}_a^x, \tilde{N}_b^y \right\}$$

! unavoidable noise and errors, finite statistics, irrational numbers...

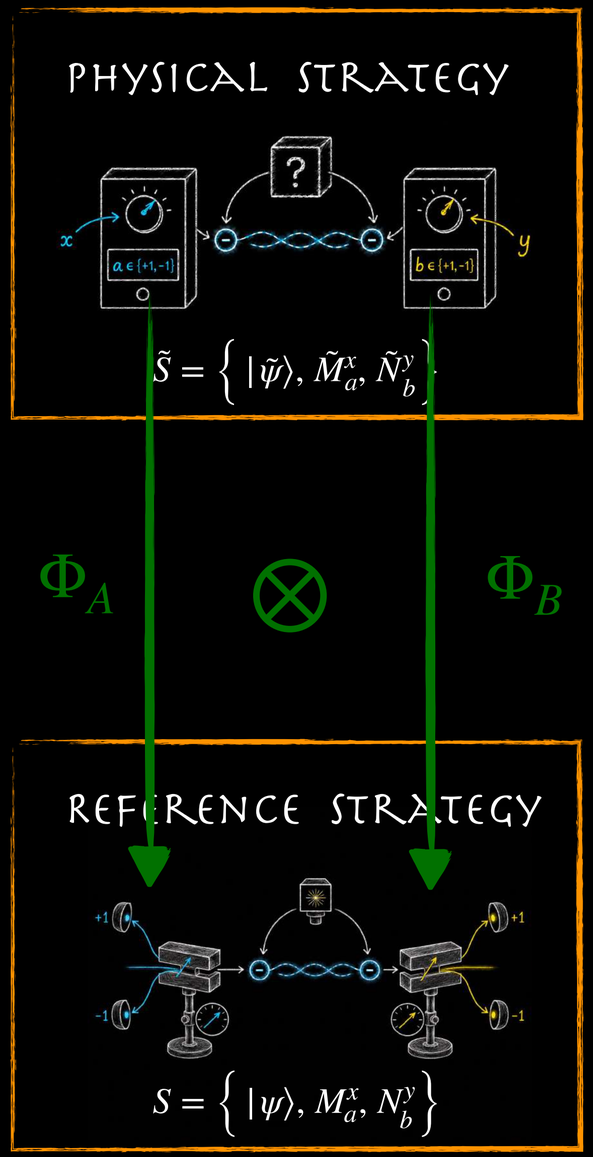
COULD LOCAL ISOMETRIES BRING PHYSICAL STRATEGY CLOSE TO THE REFERENCE ONE?



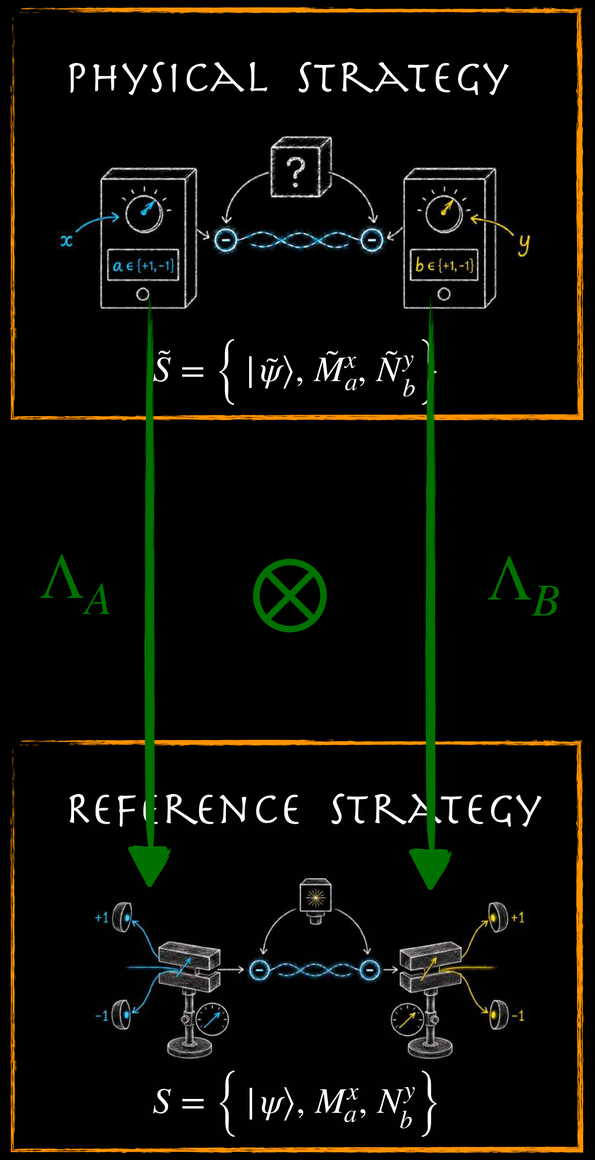
NAIVE DEFINITION: AN EVENT ROBUSTLY SELF-TESTS THE REFERENCE STRATEGY S IF EVERY PHYSICAL STRATEGY $\tilde{S}$ REPRODUCING THE EVENT UP TO A SMALL ERROR MUST HAVE A LOCAL ISOMETRY MAPPING IT CLOSE TO THE REFERENCE STRATEGY.

ROBUST SELF-TESTING

$$\Phi_A \otimes \Phi_B \left[|\tilde{\psi}\rangle \right] \approx |\psi\rangle \otimes |\text{junk}\rangle$$



ROBUST SELF-TESTING



$$\Lambda_A \otimes \Lambda_B \left[|\tilde{\psi}\rangle\right] \approx |\psi\rangle$$

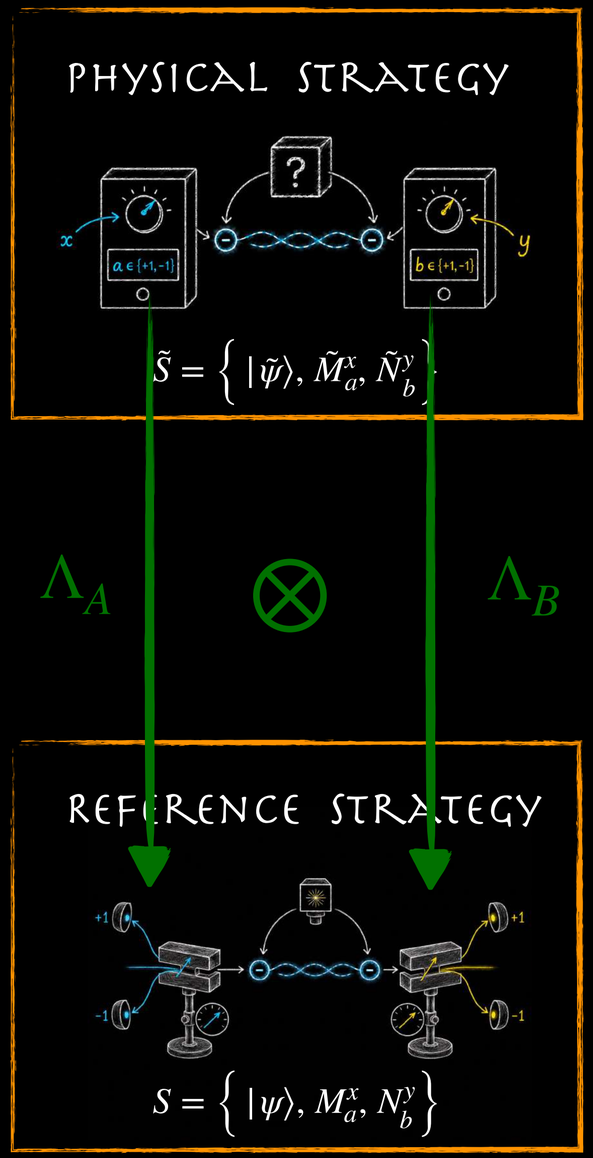
FIDELITY: $\langle\psi| \Lambda_A \otimes \Lambda_B(\tilde{\psi}) |\psi\rangle$

FIDELITY OPTIMISED OVER ALL EXTRACTION MAPS: $\max_{\Lambda_A,\Lambda_B} \langle\psi| \Lambda_A \otimes \Lambda_B(\rho) |\psi\rangle$

Kaniewski
PRL 117, 070402 [2016]

AN EVENT ROBUSTLY SELF-TESTS THE REFERENCE STATE $|\psi\rangle$ IF FROM EVERY PHYSICAL STATE $\tilde{\psi}$ REPRODUCING THE EVENT UP TO A SMALL ERROR ϵ THE EXTRACTABILITY OF THE REFERENCE STATE IS $1 - f(\epsilon)$, WHERE $\lim_{\epsilon \to 0} f(\epsilon) = 0$.

ROBUST SELF-TESTING



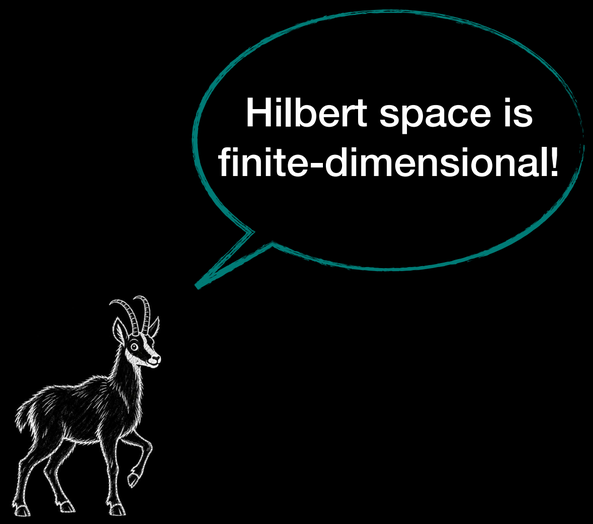
SCORE: $\omega_q - \epsilon$

MINIMIZE EXTRACTABILITY:

$$\min_{\tilde{\psi}:\omega(\tilde{\psi})=\omega_q-\epsilon} \max_{\Lambda_A,\Lambda_B} \langle\psi| \Lambda_A \otimes \Lambda_B(\rho) |\psi\rangle$$

Kaniewski
PRL 117, 070402 [2016]

ASSUMPTIONS IN SELF-TESTING



Baptista, Chen, Kaniewski, Lolck, Mančinska, Nielsen, Schmidt
Annales Henri Poincaré, [2025]

ASSUMPTIONS IN SELF-TESTING



Hilbert space is finite-dimensional!



In black-box scenarios we can always use Neimark dilation of POVMs

Baptista, Chen, Kaniewski, Lolck, Mančinska, Nielsen, Schmidt
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Baptista, Chen, Kaniewski, Lolck, Mančinska, Nielsen, Schmidt
Annales Henri Poincaré, [2025]

ASSUMPTIONS IN SELF-TESTING





LET $\mathscr{C}$ BE A CLASS OF QUANTUM STRATEGIES. AN EVENT $\mathscr{C}$-SELF-TESTS THE REFERENCE STRATEGY S IF EVERY PHYSICAL STRATEGY $\tilde{S}$ REPRODUCING THE EVENT AND BELONGING TO $\mathscr{C}$ MUST BE RELATED TO THE REFERENCE STRATEGY S THROUGH A LOCAL ISOMETRY.



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Hilbert space is finite-dimensional!

For most applications it is a fair assumption. When not we can work with self-testing in the commuting operator model !



ASSUMPTIONS IN SELF-TESTING



Hilbert space is finite-dimensional!



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Valid, if local isometries/ extraction channels act trivially on the purification Hilbert space.

ASSUMPTIONS IN SELF-TESTING



Hilbert space is finite-dimensional!



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We work with IID sources!



For most applications it is a fair assumption. When not we can work with self-testing in the commuting operator model !



Valid, if local isometries/ extraction channels act trivially on the purification Hilbert space.



For cryptographic applications we need to generalise the result to non-IID sources.

ASSUMPTIONS IN SELF-TESTING



To get my self-testing statement I assumed the state is pure, measurements are PVM and the state has full rank

I might get something from there !



Baptista, Chen, Kaniewski, Lolck, Mančinska, Nielsen, Schmidt
Annales Henri Poincaré, [2025]

ASSUMPTIONS IN SELF-TESTING

To get my self-testing statement I assumed the state is pure, measurements are PVM and the state has full rank



I might get something from there !



Could you remove either a PVM or a full rank assumption?



Actually I can, not both, but one of them I can choose!





Baptista, Chen, Kaniewski, Lolck, Mančinska, Nielsen, Schmidt
Annales Henri Poincaré, [2025]

EPR SELF-TESTING

$$\omega = \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle$$

$$\omega_c = 2 \qquad \omega_q = 2\sqrt{2}$$

CHSH INEQUALITY (GAME) IS RIGID: THE MAXIMAL VALUE (ROBUSTLY) SELF-TESTS EPR PAIR AND ANTICOMMUTING MEASUREMENTS PERFORMED BY BOTH ALICE AND BOB WITHOUT ASSUMPTIONS!

STATE SELF-TESTING

ALL BIPARTITE STATES

Coladangelo, Goh, Scarani
Nat. Comm. 8, 15485 [2017]

$$\alpha_0|00\rangle + \alpha_1|11\rangle + \alpha_2|22\rangle + \alpha_3|33\rangle$$

STATE SELF-TESTING

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Coladangelo, Goh, Scarani
Nat. Comm. 8, 15485 [2017]

$$\alpha_0|00\rangle + \alpha_1|11\rangle + \alpha_2|22\rangle + \alpha_3|33\rangle$$

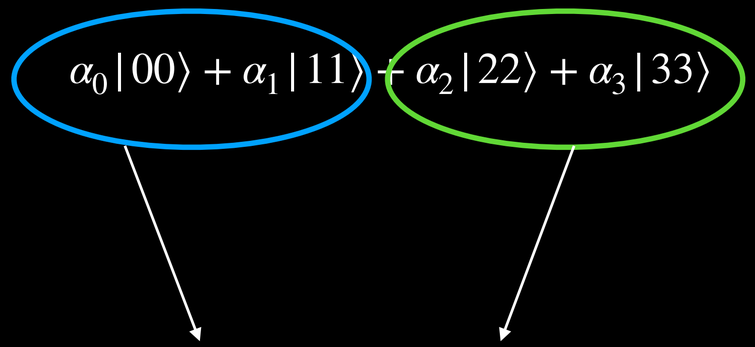
Self-testing partially entangled states
through tilted CHSH inequalities

Bamps, Pironio
PRA 91,5 [2015]

STATE SELF-TESTING

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Nat. Comm. 8, 15485 [2017]



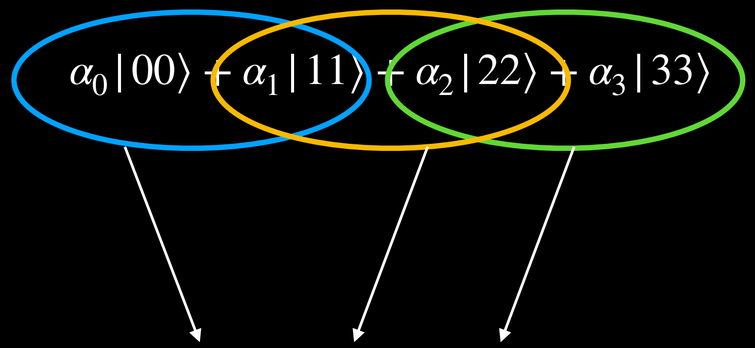
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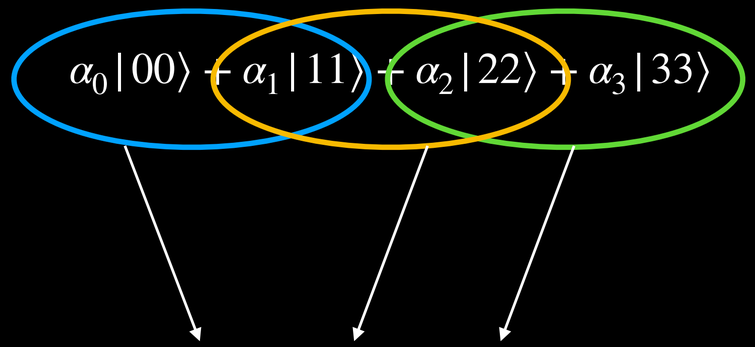
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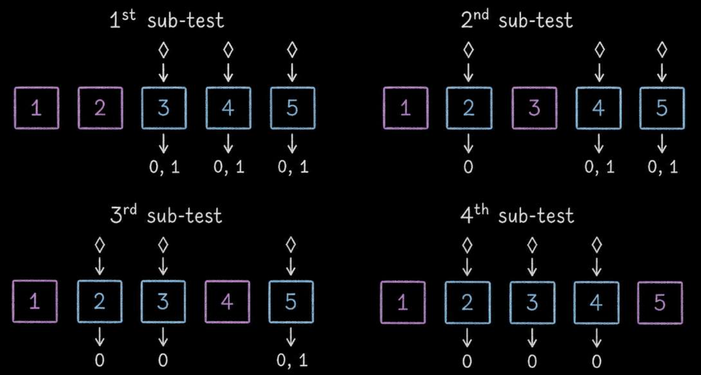


Self-testing partially entangled states through tilted CHSH inequalities

Bamps, Pironio
PRA 91,5 [2015]

ALL QUBIT MULTIPARTITE STATES

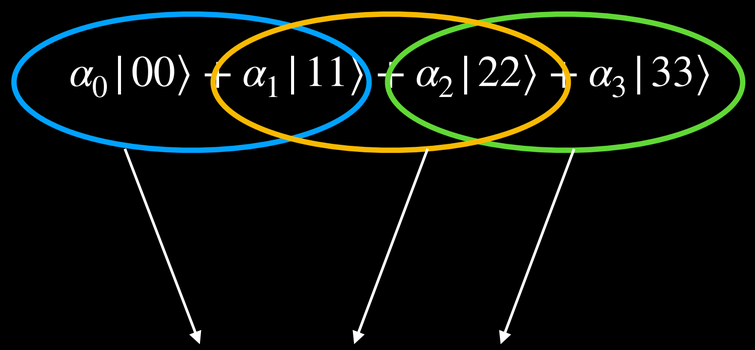
Balanzo-Juando, Coladangelo, Augusiak, Acin, Šupić
Nat. Comm. [2026]



STATE SELF-TESTING

ALL BIPARTITE STATES

Coladangelo, Goh, Scarani
Nat. Comm. 8, 15485 [2017]

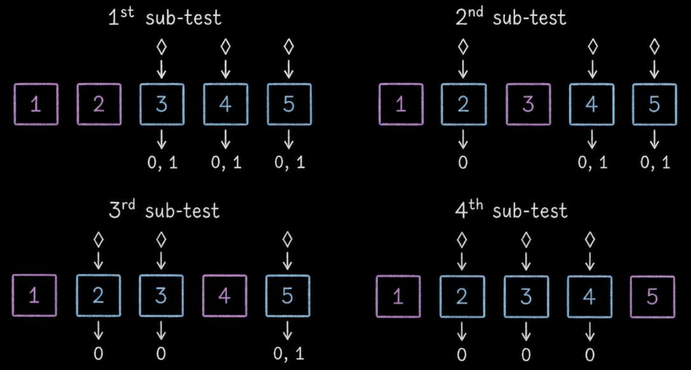


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PRA 91,5 [2015]

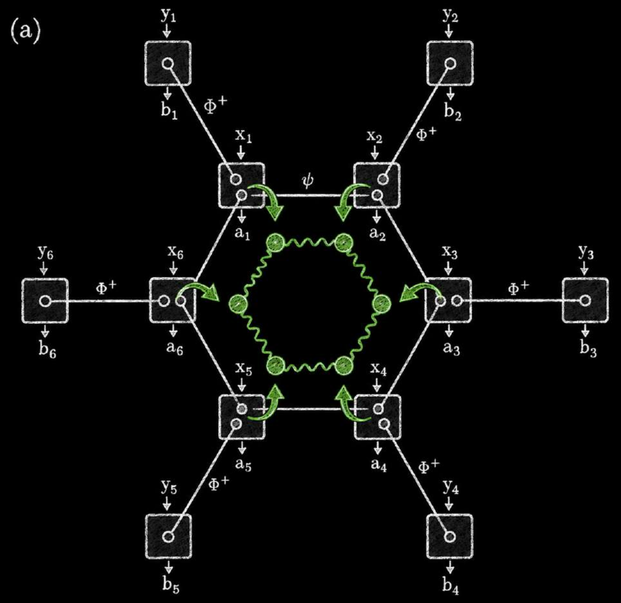
ALL QUBIT MULTIPARTITE STATES

Balanzo-Juando, Coladangelo, Augusiak, Acin, Šupić
Nat. Comm. [2026]



ALL STATES IN NETWORKS

Šupić, Bowles, Renou, Acin, Hoban
Nat. Phys. 19,5 [2023]



★ ACT II : MAIN TECHNIQUES

CHSH RIGIDITY: SUMMERS & WERNER

- a_0 and a_1 self-adjoint elements of Alice's C* algebra
- b_0 and b_1 self-adjoint elements of Bob's C* algebra
- They commute (no tensor product a priori)
- A state σ just assigns expectation value to

$$\chi = \frac{1}{2}\left(a_0(b_0 + b_1) + a_1(b_0 - b_1)\right)$$

- GNS representation gives us a Hilbert space, and a unit vector inside it $|\Omega\rangle$, and representations of self-adjoint elements: A_0, A_1, B_0, B_1
- $$A = \frac{1}{2}\left(A_0 + iA_1\right),$$
  $$B = \frac{1}{2\sqrt{2}}\left(B_0 + B_1 + i(B_0 - B_1)\right)$$
- $$AA^\dagger + A^\dagger A = \frac{1}{2}\left(A_0^2 + A_1^2\right) \leq 1,$$
  $$BB^\dagger + B^\dagger B = \frac{1}{2}\left(B_0^2 + B_1^2\right) \leq 1$$

Summers, Werner, 1987
J. Math. Phys. 28,2440

CHSH RIGIDITY: SUMMERS & WERNER

- $\sqrt{2}\sigma(\chi) = 4\mathrm{Re}\langle\Omega|A^\dagger B|\Omega\rangle$
- Since quantum value is $2\sqrt{2}$, and norm of $A|\Omega\rangle$ and $B|\Omega\rangle$ are at most 1, it must be $A|\Omega\rangle = B|\Omega\rangle$, and A_i and B_i are sharp observables
- Since A_i and B_j commute we have $AA|\Omega\rangle = BB|\Omega\rangle$

- This reduces to $(A_0A_1 + A_1A_0)|\Omega\rangle = (B_0^2 - B_1^2)|\Omega\rangle$
- As B_0 and B_1 are sharp, A_0 and A_1 anticommute on the support of the state
- Symmetrical analysis shows that Bob's observables also anti-commute on the support of the state.

Summers, Werner, 1987
J. Math. Phys. 28,2440

CHSH RIGIDITY: TSIRELSON

- Given
  $[A_k, B_l] = 0, \ \| A_k \| , \ \| B_l \| \ \leq 1$
  , density matrix W,
  $c_{kl} = \mathrm{Tr}(A_k B_l W)$,
  $\beta = c_{00} + c_{01} + c_{10} - c_{11}$

- Vectorise:
  $\langle X, Y \rangle_W = \mathrm{Re}\ \mathrm{Tr}(Y^* X W)$: the observables become vectors,
  $\|A_k\|_W, \|B_l\|_W \leq 1$,
  $\langle A_k, B_l \rangle_W = c_{kl}$

- Bound=geometry:
  $\beta = \langle A_0, B_0 + B_1 \rangle + \langle A_1, B_0 - B_1 \rangle \leq$
  $\leq \sqrt{2}(\|B_0 + B_1\|^2 + \|B_0 - B_1\|^2)^{1/2}$
  $\beta \leq \sqrt{2}(2\|B_0\|^2 + 2\|B_1\|^2)^{1/2} \leq 2\sqrt{2}$,
  Cauchy-Schwarz + parallelogram inequalities

- Saturation gives rigid geometry:
  $\beta = 2\sqrt{2}$ forces two orthonormalisation frames at 45 degrees: $A_k^2 W = W$,
  $\sqrt{2} A_1 W = (B_0 + B_1) W$, ...,
  $\{A_0, A_1\} W = 0, \ \ \{B_0, B_1\} W = 0$

Tsirelson, 1987
J. Sov. Math.

CHSH RIGIDITY: TSIRELSON

- From non degeneracy to operator inequalities: B-monomials on $\mathrm{ran} W$ span $\mathscr{H}$, so $A_k^2 = \mathbb{I}$, $B_l^2 = \mathbb{I}$, $\{A_0, A_1\} = 0$, $\{B_0, B_1\} = 0$, this is a representation of two Clifford algebras
- Representation theory: two anticommuting involutions are unitarily equivalent to $\sigma_z \otimes \mathbb{I}$ and $\sigma_x \otimes \mathbb{I}$, and Hilbert space is isomorphic to $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathscr{R}$

- State: the top eigenvalue of $\sqrt{2}(\sigma_z \otimes \sigma_z + \sigma_x \otimes \sigma_x) \otimes \mathbb{I}$ is $2\sqrt{2}$ and the corresponding eigenvector is $|\Phi_+\rangle$

Tsirelson, 1987
J. Sov. Math.

CHSH RIGIDITY: SOS DECOMPOSITIONS

- Shifted Bell operator

$$4\mathbb{I} - \sqrt{2}(A_0B_0 + A_0B_1 + A_1B_0 - A_1B_1)$$
$$= \left(A_0 - \frac{B_0 + B_1}{\sqrt{2}}\right)^2 + \left(A_1 - \frac{B_0 - B_1}{\sqrt{2}}\right)^2$$

- $4 - \sqrt{2}\beta = \langle\psi| \left(A_0 - \frac{B_0 + B_1}{\sqrt{2}}\right)^2 |\psi\rangle + \langle\psi| \left(A_1 - \frac{B_0 - B_1}{\sqrt{2}}\right)^2 |\psi\rangle$

- For a state $|\psi\rangle$ reaching the maximal violation we thus have $\left(A_0 - \frac{B_0 + B_1}{\sqrt{2}}\right)|\psi\rangle = 0, \quad \left(A_1 - \frac{B_0 - B_1}{\sqrt{2}}\right)|\psi\rangle = 0$

- $A_0|\psi\rangle = \frac{B_0 + B_1}{\sqrt{2}}|\psi\rangle, \quad A_1|\psi\rangle = \frac{B_0 - B_1}{\sqrt{2}}|\psi\rangle$

- $(A_0A_1 + A_1A_0)|\psi\rangle = 0$

- $(B_0B_1 + B_1B_0)|\psi\rangle = 0$

CHSH RIGIDITY: SWAP ISOMETRY

$$(A_0A_1 + A_1A_0)\,|\psi\rangle = 0$$

$$(B_0B_1 + B_1B_0)\,|\psi\rangle = 0$$

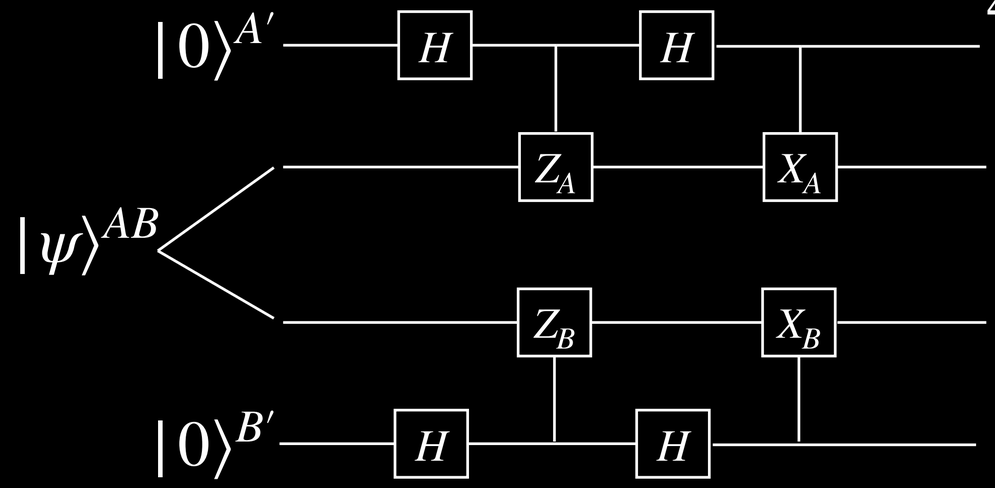
$$A_0\,|\psi\rangle = \frac{B_0 + B_1}{\sqrt{2}}\,|\psi\rangle, \quad A_1\,|\psi\rangle = \frac{B_0 - B_1}{\sqrt{2}}\,|\psi\rangle$$

$$Z_A = A_0, \quad X_A = A_1$$

$$Z_B = \frac{B_0 + B_1}{\sqrt{2}}, \quad X_B = \frac{B_0 - B_1}{\sqrt{2}}$$

$$Z_A\,|\psi\rangle = Z_B\,|\psi\rangle, \qquad X_A\,|\psi\rangle = X_B\,|\psi\rangle$$

Directly construct the SWAP isometry



$$4SWAP(|00\rangle^{A'B'}\,|\psi\rangle^{AB}) = |00\rangle^{A'B'}(\mathbb{I} + Z_A)(\mathbb{I} + Z_B)\,|\psi\rangle^{AB}$$

$$+\,|10\rangle^{A'B'}X_A(\mathbb{I} - Z_A)(\mathbb{I} + Z_B)\,|\psi\rangle^{AB}$$

$$+\,|01\rangle^{A'B'}(\mathbb{I} + Z_A)X_B(\mathbb{I} - Z_B)\,|\psi\rangle^{AB}$$

$$+\,|11\rangle^{A'B'}X_A(\mathbb{I} - Z_A)X_B(\mathbb{I} - Z_B)\,|\psi\rangle^{AB}$$

CHSH RIGIDITY: SWAP ISOMETRY

$$(\mathbb{I} + Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = (\mathbb{I} + Z_A)(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$(\mathbb{I} + Z_A)(\mathbb{I} + Z_B)\,|\psi\rangle = 2(\mathbb{I} + Z_A)\,|\psi\rangle$$

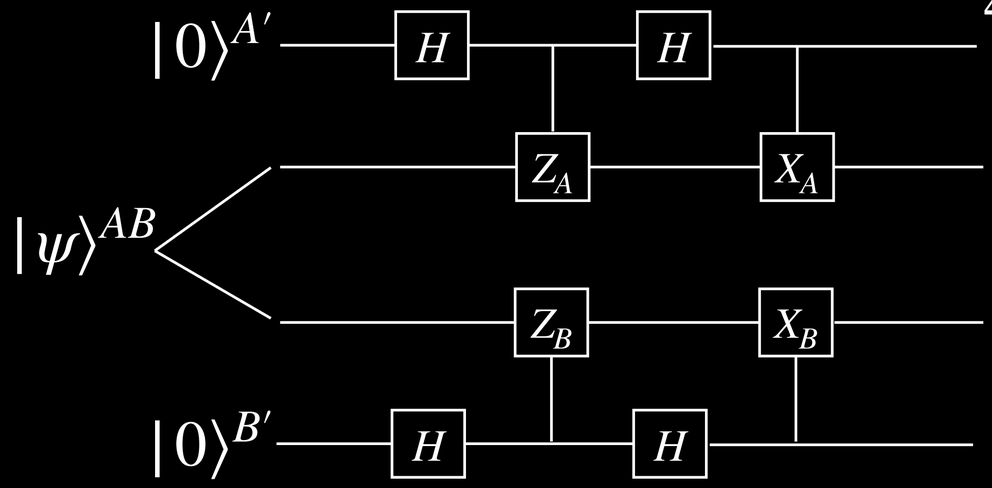
$$(\mathbb{I} - Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = 2(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$Z_A = A_0, \quad X_A = A_1$$

$$Z_B = \frac{B_0 + B_1}{\sqrt{2}}, \quad X_B = \frac{B_0 - B_1}{\sqrt{2}}$$

$$Z_A\,|\psi\rangle = Z_B\,|\psi\rangle, \qquad X_A\,|\psi\rangle = X_B\,|\psi\rangle$$

Directly construct the SWAP isometry



$$4SWAP(|00\rangle^{A'B'}\,|\psi\rangle^{AB}) = 2\,|00\rangle^{A'B'}(\mathbb{I} + Z_A)\,|\psi\rangle^{AB}$$

$$+0$$

$$+0$$

$$+2\,|11\rangle^{A'B'} X_A X_B (\mathbb{I} - Z_A)\,|\psi\rangle^{AB}$$

CHSH RIGIDITY: SWAP ISOMETRY

$$(\mathbb{I} + Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = (\mathbb{I} + Z_A)(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$(\mathbb{I} + Z_A)(\mathbb{I} + Z_B)\,|\psi\rangle = 2(\mathbb{I} + Z_A)\,|\psi\rangle$$

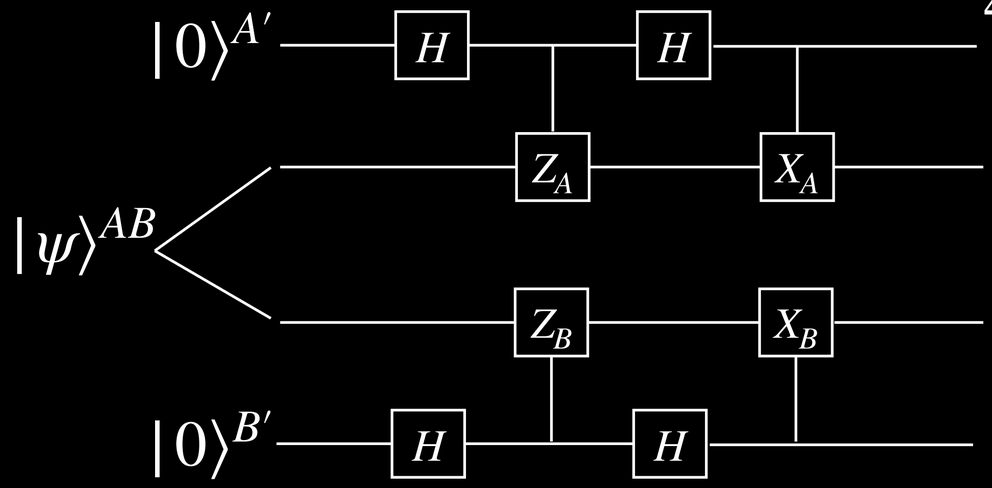
$$(\mathbb{I} - Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = 2(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$Z_A = A_0, \quad X_A = A_1$$

$$Z_B = \frac{B_0 + B_1}{\sqrt{2}}, \quad X_B = \frac{B_0 - B_1}{\sqrt{2}}$$

$$Z_A\,|\psi\rangle = Z_B\,|\psi\rangle, \qquad X_A\,|\psi\rangle = X_B\,|\psi\rangle$$

Directly construct the SWAP isometry



$$4SWAP(|00\rangle^{A'B'}\,|\psi\rangle^{AB}) = 2\,|00\rangle^{A'B'}(\mathbb{I} + Z_A)\,|\psi\rangle^{AB}$$

$$+0$$

$$+0$$

$$+2\,|11\rangle^{A'B'}X_A(\mathbb{I} - Z_A)X_A\,|\psi\rangle^{AB}$$

CHSH RIGIDITY: SWAP ISOMETRY

$$(\mathbb{I} + Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = (\mathbb{I} + Z_A)(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$(\mathbb{I} + Z_A)(\mathbb{I} + Z_B)\,|\psi\rangle = 2(\mathbb{I} + Z_A)\,|\psi\rangle$$

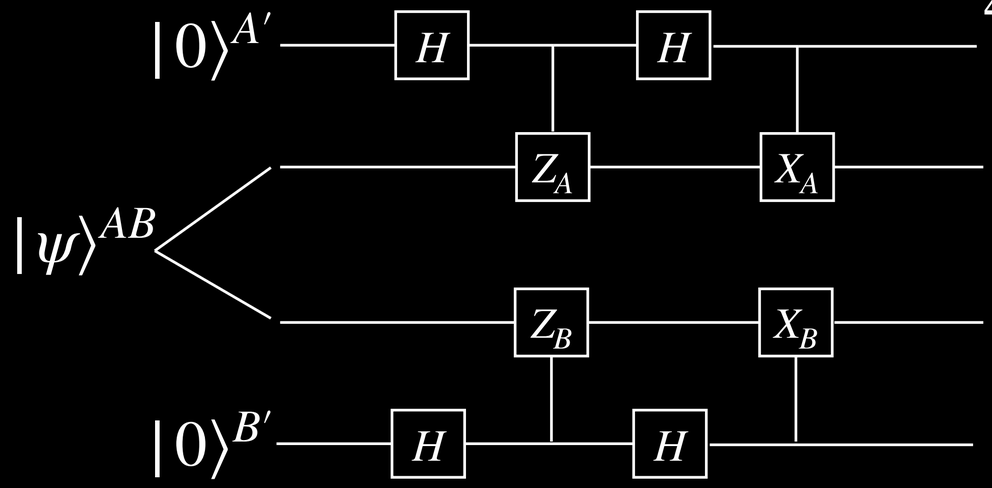
$$(\mathbb{I} - Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = 2(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$Z_A = A_0, \quad X_A = A_1$$

$$Z_B = \frac{B_0 + B_1}{\sqrt{2}}, \quad X_B = \frac{B_0 - B_1}{\sqrt{2}}$$

$$Z_A\,|\psi\rangle = Z_B\,|\psi\rangle, \qquad X_A\,|\psi\rangle = X_B\,|\psi\rangle$$

Directly construct the SWAP isometry



$$4SWAP(|00\rangle^{A'B'}\,|\psi\rangle^{AB}) = 2\,|00\rangle^{A'B'}(\mathbb{I} + Z_A)\,|\psi\rangle^{AB}$$

$$+0$$

$$+0$$

$$+2\,|11\rangle^{A'B'}(\mathbb{I} + Z_A)X_AX_A\,|\psi\rangle^{AB}$$

CHSH RIGIDITY: SWAP ISOMETRY

$$(\mathbb{I} + Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = (\mathbb{I} + Z_A)(\mathbb{I} - Z_A)\,|\psi\rangle$$

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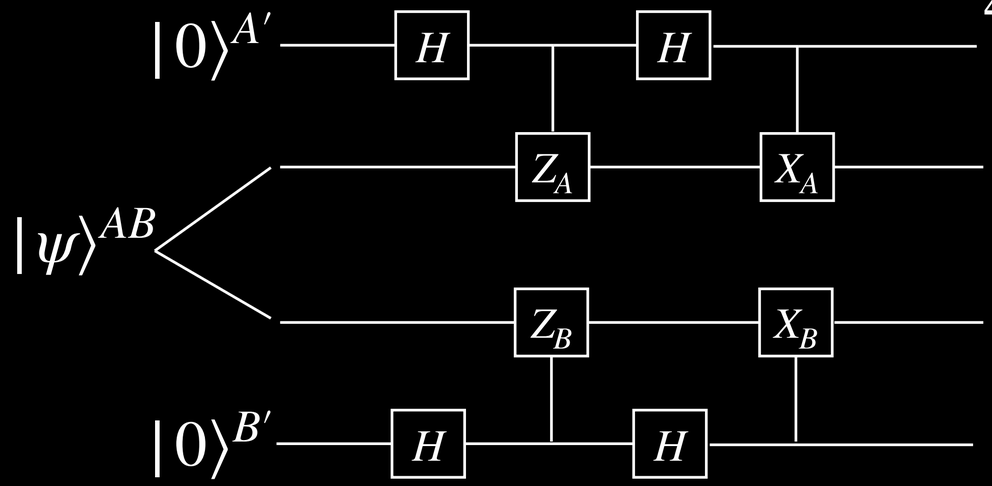
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Directly construct the SWAP isometry



$$4SWAP(|00\rangle^{A'B'}\,|\psi\rangle^{AB}) = 2\,|00\rangle^{A'B'}(\mathbb{I} + Z_A)\,|\psi\rangle^{AB}$$

$$+0$$

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$$+2\,|11\rangle^{A'B'}(\mathbb{I} + Z_A)\,|\psi\rangle^{AB}$$

CHSH RIGIDITY: SWAP ISOMETRY

$$(\mathbb{I} + Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = (\mathbb{I} + Z_A)(\mathbb{I} - Z_A)\,|\psi\rangle$$

$$(\mathbb{I} + Z_A)(\mathbb{I} + Z_B)\,|\psi\rangle = 2(\mathbb{I} + Z_A)\,|\psi\rangle$$

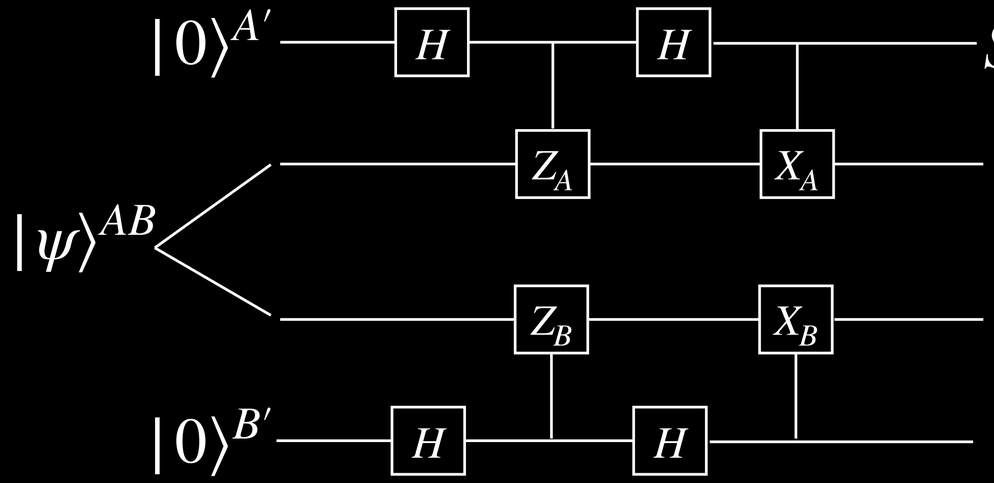
$$(\mathbb{I} - Z_A)(\mathbb{I} - Z_B)\,|\psi\rangle = 2(\mathbb{I} - Z_A)\,|\psi\rangle$$

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$$Z_A\,|\psi\rangle = Z_B\,|\psi\rangle, \qquad X_A\,|\psi\rangle = X_B\,|\psi\rangle$$

Directly construct the SWAP isometry

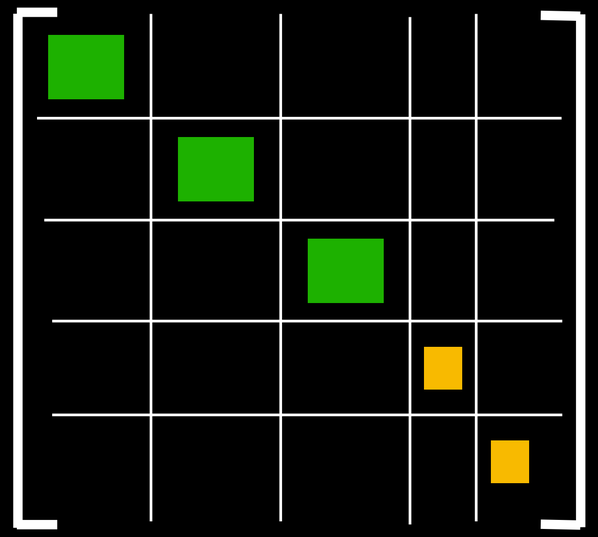


$$SWAP(|00\rangle^{A'B'}\,|\psi\rangle^{AB}) = \frac{|00\rangle^{A'B'} + |11\rangle^{A'B'}}{\sqrt{2}} \otimes \frac{\mathbb{I} + Z_A}{\sqrt{2}}\,|\psi\rangle^{AB}$$

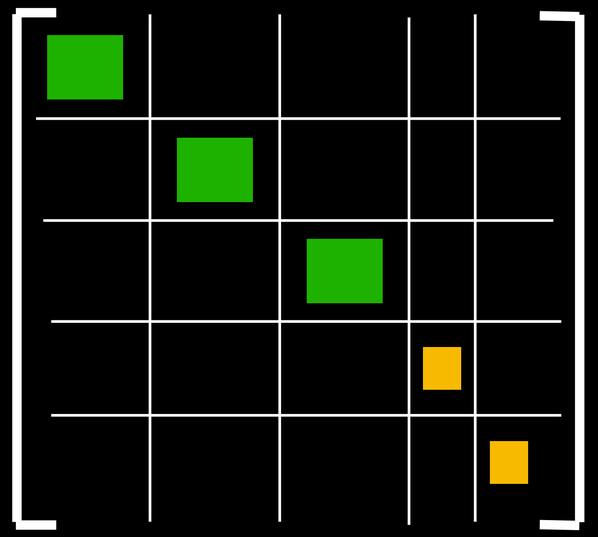
$$= |\mathrm{EPR}\rangle^{AB} \otimes |\mathrm{aux}\rangle^{A'B'}$$

CHSH RIGIDITY: JORDAN'S LEMMA

$VA_0V^\dagger =$



$VA_1V^\dagger =$



A few steps and we get result: For any two anti-commuting observables X and Z there is a unitary U such that

$$UZU^\dagger = \sigma_z \otimes \mathbb{I}. \qquad UXU^\dagger = \sigma_x \otimes \mathbb{I}$$

Applying to Alice's and Bob's observables we thus obtain

$$UA_0U^\dagger = \sigma_z \otimes \mathbb{I} \qquad UA_1U^\dagger = \sigma_x \otimes \mathbb{I}$$

$$UB_0U^\dagger = \frac{\sigma_z + \sigma_x}{\sqrt{2}} \otimes \mathbb{I} \qquad UB_1U^\dagger = \frac{\sigma_z - \sigma_x}{\sqrt{2}} \otimes \mathbb{I}.$$

So the Bell operator then is $\sqrt{2}(\sigma_z \otimes \sigma_z + \sigma_x \otimes \sigma_x) \otimes \mathbb{I} \otimes \mathbb{I}$ and the eigenstate corresponding to the largest eigenvalue ($2\sqrt{2}$) is the EPR pair

MAYERS-YAO SELF-TEST

- Mayers-Yao:
  $\langle\psi|A_0B_0|\psi\rangle = 1,\ \langle\psi|A_0B_1|\psi\rangle = 0,\ \langle\psi|A_0B_2|\psi\rangle = 1/\sqrt{2},$
  $\langle\psi|A_1B_0|\psi\rangle = 0,\ \langle\psi|A_1B_1|\psi\rangle = 1,\ \langle\psi|A_1B_2|\psi\rangle = 1/\sqrt{2},$

  $\vec{a}_k = A_k|\psi\rangle,\ \vec{b}_l = B_l|\psi\rangle,$

  $\langle\vec{a}_0,\vec{b}_0\rangle = 1,\ \langle\vec{a}_1,\vec{b}_1\rangle = 1,\ \langle\vec{a}_0,\vec{b}_1\rangle = 0,\ \langle\vec{a}_1,\vec{b}_0\rangle = 0$

$$\langle\vec{a}_0,\vec{b}_2\rangle = \langle\vec{a}_1,\vec{b}_2\rangle = \frac{1}{\sqrt{2}}$$

Mayers, Yao
QIC 4,4 [2004]

MAYERS-YAO SELF-TEST

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$$\langle\vec{a}_0,\vec{b}_2\rangle = \langle\vec{a}_1,\vec{b}_2\rangle = \frac{1}{\sqrt{2}}$$

$\vec{a}_0 = \vec{b}_0$

Mayers, Yao
QIC 4,4 [2004]

MAYERS-YAO SELF-TEST

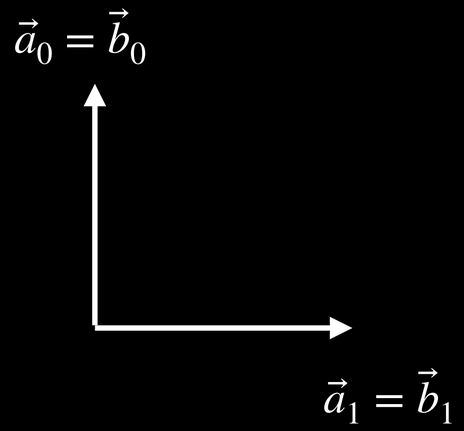
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$$\langle\vec{a}_0, \vec{b}_2\rangle = \langle\vec{a}_1, \vec{b}_2\rangle = \frac{1}{\sqrt{2}}$$



Mayers, Yao
QIC 4,4 [2004]

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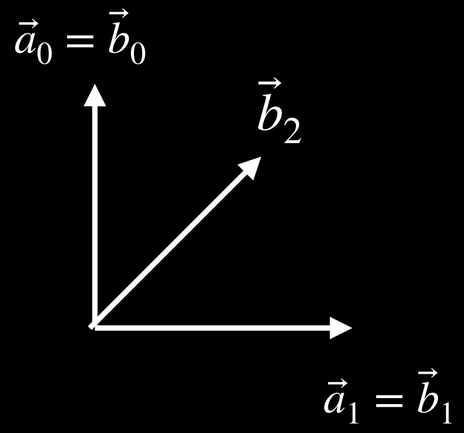
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Mayers, Yao
QIC 4,4 [2004]

MAYERS-YAO SELF-TEST

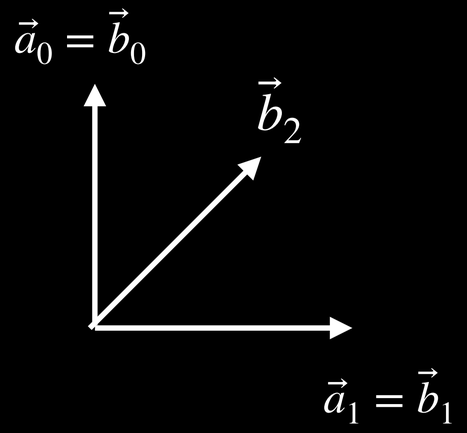
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$$\langle\vec{a}_0, \vec{b}_2\rangle = \langle\vec{a}_1, \vec{b}_2\rangle = \frac{1}{\sqrt{2}}$$



$$\vec{b}_2 = \frac{1}{\sqrt{2}}\vec{b}_0 + \frac{1}{\sqrt{2}}\vec{b}_1$$

$$B_2|\psi\rangle = \frac{B_0 + B_1}{\sqrt{2}}|\psi\rangle$$

$$B_2B_2|\psi\rangle = |\psi\rangle + \frac{\{B_0, B_1\}}{\sqrt{2}}|\psi\rangle$$

$$\{B_0, B_1\}|\psi\rangle = 0$$

Mayers, Yao
QIC 4,4 [2004]

ROBUSTNESS BOUNDS

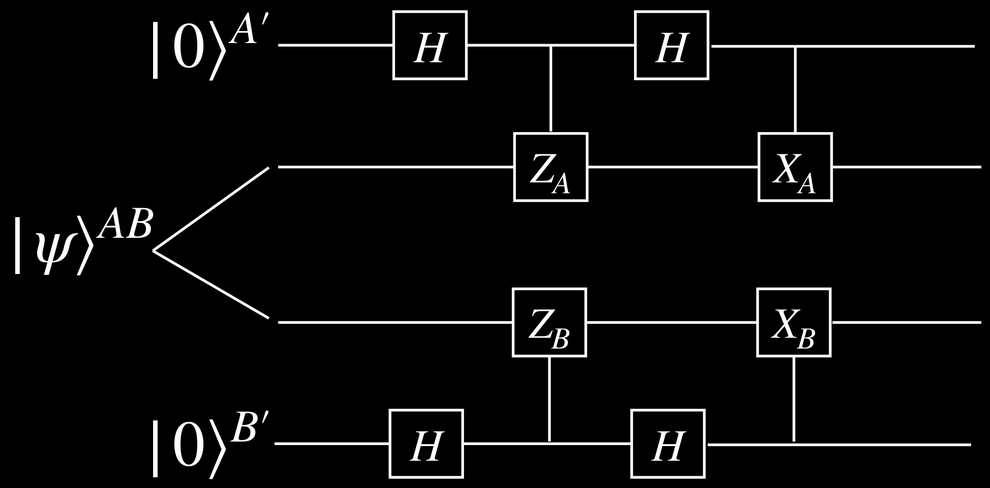
$$\min_{\rho:\beta(\rho)=2\sqrt{2}-\epsilon} \max_{\Lambda_A,\Lambda_B} \langle \mathrm{EPR} | \Lambda_A \otimes \Lambda_B(\rho) | \mathrm{EPR} \rangle$$

Yang, Vertesi, Bancal, Scarani, Navascues
PRL 113, 4 [2014]

Working with a specific isometry we get rid of maximization, but then we have a lower bound on extractability

SWAP method chooses the SWAP isometry

$$\min_{\rho:\beta(\rho)=2\sqrt{2}-\epsilon} \langle \mathrm{EPR} | SWAP(\rho) | \mathrm{EPR} \rangle$$



$$4SWAP(|00\rangle^{A'B'}|\psi\rangle^{AB}) = \sum_{ij=0}^{1} |ij\rangle^{A'B'} X_A^i(\mathbb{I} + (-1)^i Z_A) X_B^j(\mathbb{I} + (-1)^j Z_B)|\psi\rangle^{AB}$$

$$16SWAP(|00\rangle^{A'B'}|\psi\rangle^{AB}) = \sum_{ijkl=0}^{1} |ij\rangle\langle kl|^{A'B'} S_{ij}|\psi\rangle\langle\psi|^{AB} S_{kl}^{\dagger}$$

$$SWAP(\rho) = \mathrm{Tr}_{AB} 16SWAP(|00\rangle^{A'B'}|\psi\rangle^{AB}) = \sum_{ijkl=0}^{1} |ij\rangle\langle kl|^{A'B'} S_{ij}|\psi\rangle\langle\psi|^{AB} S_{kl}^{\dagger}$$

$$SWAP(\rho) = \mathrm{Tr}_{AB} 16SWAP(|00\rangle^{A'B'}|\psi\rangle^{AB}) = \sum_{ijkl=0}^{1} |ij\rangle\langle kl|^{A'B'} S_{ij}|\psi\rangle\langle\psi|^{AB} S_{kl}^{\dagger}$$

$$SWAP(\rho) = \sum_{ijkl=0}^{1} |ij\rangle\langle kl|^{A'B'} \langle\psi|R_{ijkl}|\psi\rangle$$

$$\min_{\rho:\beta(\rho)=2\sqrt{2}-\epsilon} \langle \mathrm{EPR} | SWAP(\rho) | \mathrm{EPR} \rangle = \min_{\rho:\beta(\rho)=2\sqrt{2}-\epsilon} \sum_{ijkl=0}^{1} \langle\psi|\tilde{R}_{ijkl}|\psi\rangle$$

Use NPA hierarchy for minimization!

ROBUSTNESS BOUNDS

$$\min_{\rho:\beta(\rho)=2\sqrt{2}-\epsilon} \max_{\Lambda_A,\Lambda_B} \langle\psi|\,\Lambda_A\otimes\Lambda_B(\rho)\,|\psi\rangle$$

Kaniewski
PRL 117, 070402 [2016]

Bell operator $\mathscr{B}$ and Bell value $\mathrm{Tr}[\mathscr{B}\psi]$

Reference state: $\tilde{\psi}$

Operator inequality $\Lambda_A^\dagger\otimes\Lambda_B^\dagger(\tilde{\psi})\geq s\mathscr{B}+\mu$ implies $\Xi\geq s\beta+\mu$

1. Apply Jordan's lemma
2. Choose extraction maps (some kind of dephasing channel)
3. Prove the operator inequality!

REPRESENTATION THEORY APPROACH

GOWERS-HATAMI THEOREM

$$X^2|\psi\rangle = \mathbb{I}|\psi\rangle \quad Z^2|\psi\rangle = \mathbb{I}|\psi\rangle \quad \{X,Z\}|\psi\rangle = 0$$

Pauli group

$$X^2 = \mathbb{I} \quad Z^2 = \mathbb{I}; \quad \{X,Z\} = 0 \qquad \xleftarrow{\pi} \qquad x,\ z,\ e,\ x^2 = e,\ z^2 = e,\ xz = -xz$$

$$\|X^2|\psi\rangle - |\psi\rangle\| \leq \epsilon$$

$$\|Z^2|\psi\rangle - |\psi\rangle\| \leq \epsilon$$

$$\|XZ|\psi\rangle + ZX|\psi\rangle\| \leq \epsilon$$

Is there a representation of the Pauli group such that X and Z are close to it?

Gowers, Hatami
Sbor. Math. 208 (12), [2017]

GOWERS-HATAMI THEOREM

JOURNAL OF COMPUTER AND SYSTEM SCIENCES **47**, 549–595 (1993)

Self-Testing/Correcting
with Applications to Numerical Problems

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MICHAEL LUBY†

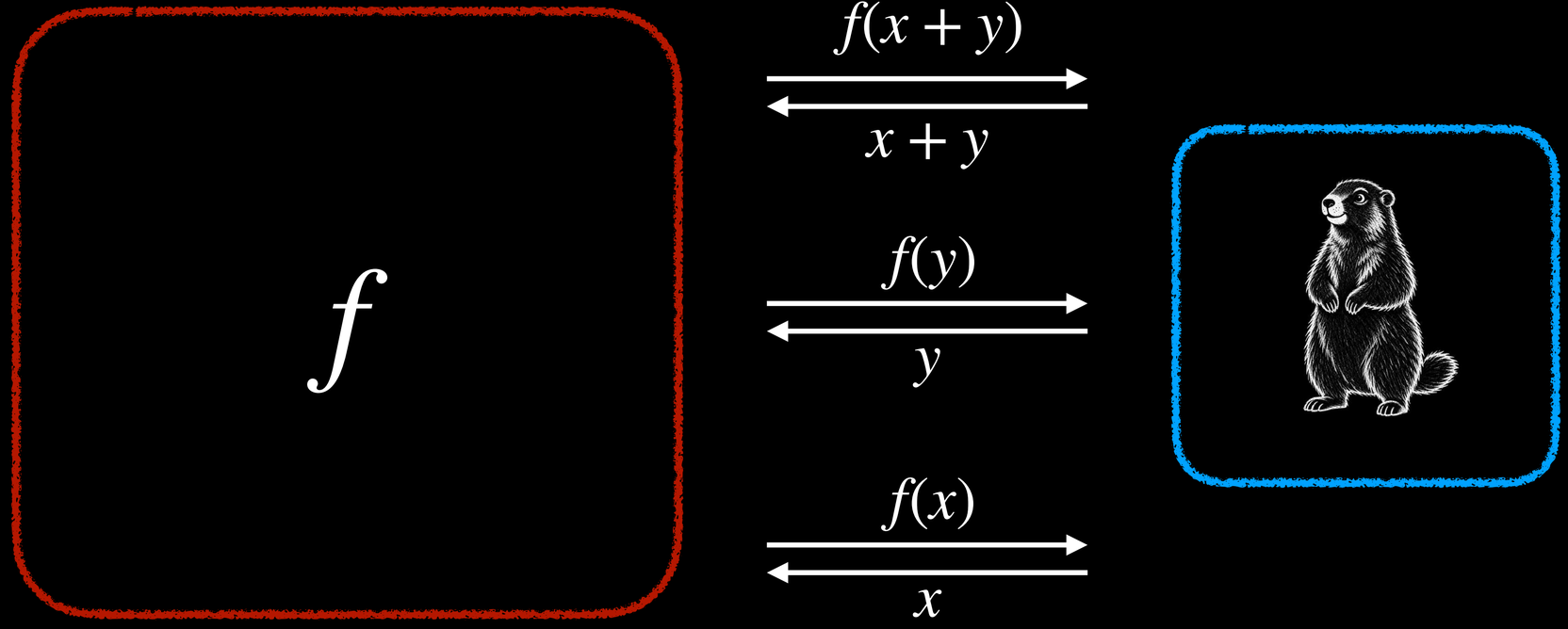
International Computer Science Institute, Berkeley, California 94704;
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Received August 14, 1990; revised November 19, 1991

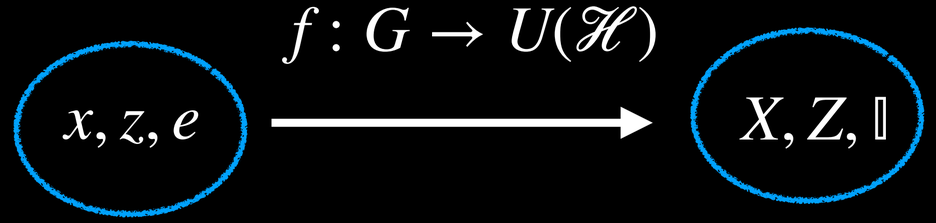


Passing the test implies that the function is close to some linear function

GOWERS-HATAMI THEOREM

STABILITY OF APPROXIMATE REPRESENTATIONS

$$x, z, e, x^2 = e, z^2 = e, xz = -xz$$



$$\|X^2|\psi\rangle - |\psi\rangle\| \leq \epsilon$$

$$\|Z^2|\psi\rangle - |\psi\rangle\| \leq \epsilon$$

$$\|XZ|\psi\rangle + ZX|\psi\rangle\| \leq \epsilon$$

∃ isometry $V : \mathcal{H} \to \mathcal{K}$
∃ representation ϕ such that
$\|Vf(g) - \phi(g)V\|_\psi^2 \leq \sqrt{\epsilon}$

$$V^*\phi(g)V = \mathbb{E}_{h \in G} f(h)^* f(hg)$$

Gowers, Hatami
Sbor. Math. 208 (12), [2017]

ALGEBRAIC APPROACH

C* ALGEBRAS

C* algebra = Banach algebra over a field of complex numbers with a map $x \to x^*$

A. $x^{**} = x$

B. $(x+y)^* = x^* + y^* \quad (xy)^* = y^* x^*$

C. $(\lambda x)^* = \bar{\lambda} x^*$

D. $\|xx* \| = \|x\| \|x^*\|$

Example: algebra of bounded operators on a Hilbert space

State on the algebra: a unit positive functional $\phi : \mathcal{A} \to \mathbb{C}$

Representation: a homomorphism $\pi : \mathcal{A} \to B(\mathcal{H})$

GNS representation:

Every state ϕ induces a representation $\pi_\phi : \mathcal{A} \to B(\mathcal{H}_\phi)$ and a unit cyclic vector $|\Omega_\phi\rangle$, such that $\phi(\mathfrak{a}) = \langle \Omega_\phi | \pi_\phi(\mathfrak{a}) | \Omega_\phi \rangle$

Cyclic: $\overline{\pi(\mathcal{A})} |\Omega_\phi\rangle = \mathcal{H}_\phi$

Universal C* algebras

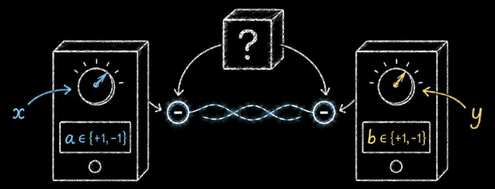
Universal C* algebra

Fix set of generators g and a relations between them r, run over *every* concrete realization of the relations and take the largest operator norm that ever occurs.

Universality property

For every instant of operators satisfying the relations, there is a representation from the universal C* algebra to the concrete algebra, such that the generators are mapped to the operators satisfying relations

UNIVERSAL POVM ALGEBRAS



Alice's POVMs: M_a^x

Bob's POVMs: N_b^y

Shared state: ρ

Define generators $\mathfrak{m}_a^x$, $\mathfrak{n}_b^y$

Choose relations $\mathfrak{m}_a^x, \mathfrak{n}_b^y \geq 0, \sum_a \mathfrak{m}_a^x = 1, \sum_b \mathfrak{n}_b^y = 1$

Universal POVM algebra: $\mathcal{A}_A \otimes_{min} \mathcal{A}_B$

Universality property: There is a representation π such that $\pi(\mathfrak{m}_a^x \otimes \mathfrak{n}_b^y) = M_a^x \otimes N_b^y$

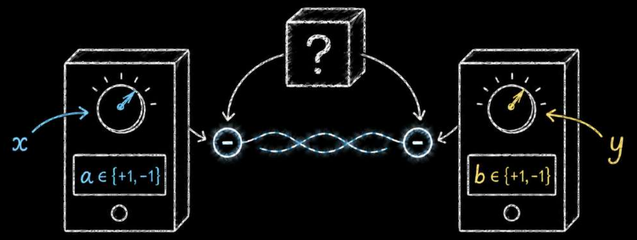
Shared state ρ induces an algebraic state ϕ_ρ, such that $\phi_\rho(\mathfrak{a}) = \mathrm{Tr}[\rho\pi(\mathfrak{a})]$

Correlations are obtained as this state evaluated on generators

$\phi_\rho(\mathfrak{m}_a^x \otimes \mathfrak{n}_b^y) = \mathrm{Tr}[\rho M_a^x \otimes N_b^y] = p(a, b \,|\, x, y)$

ABSTRACT SELF-TESTING

Paddock, Slofstra, Zhao, Zhou
AHP 25(10), [2024]



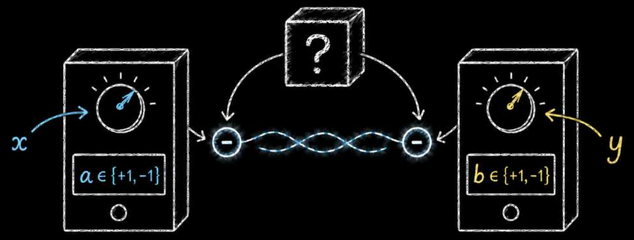
Alice's POVMs: M_a^x "Alice's" generators $\mathfrak{m}_a^x$

Bob's POVMs: N_b^y "Bob's" generators $\mathfrak{n}_b^y$

Shared state: ρ Algebraic state ϕ_ρ

LET S BE A SUBSET OF STATES ON THE UNIVERSAL POVM ALGEBRA. AN EVENT IS AN ABSTRACT SELF-TEST FOR S IF THERE IS A UNIQUE STATE IN S COMPATIBLE WITH THE EVENT.

ABSTRACT SELF-TESTING



Paddock, Slofstra, Zhao, Zhou
AHP 25(10), [2024]

$$|\psi\rangle \quad \{M_a^x, N_b^y\} \xleftarrow{\pi} \{\mathfrak{m}_a^x, \mathfrak{n}_b^y\} \xrightarrow{\bar{\pi}} \{\bar{M}_a^x, \bar{N}_b^y\} \quad |\bar{\psi}\rangle$$

$$\phi_\psi(\Pi(\mathfrak{m}_a^x \otimes \mathfrak{n}_b^y)) = \langle\psi|\Pi\left(\pi(\mathfrak{m}_a^x) \otimes \pi(\mathfrak{n}_b^y)\right)|\psi\rangle$$

$$\phi_{\bar{\psi}}(\Pi(\mathfrak{m}_a^x \otimes \mathfrak{n}_b^y)) = \langle\bar{\psi}|\Pi\left(\bar{\pi}(\mathfrak{m}_a^x) \otimes \bar{\pi}(\mathfrak{n}_b^y)\right)|\bar{\psi}\rangle$$

$$\phi_\psi(\Pi(\mathfrak{m}_a^x \otimes \mathfrak{n}_b^y)) = \langle\psi|\Pi\left(M_a^x \otimes N_b^y\right)|\psi\rangle$$

$$\phi_{\bar{\psi}}(\Pi(\mathfrak{m}_a^x \otimes \mathfrak{n}_b^y)) = \langle\bar{\psi}|\Pi\left(\bar{M}_a^x \otimes \bar{N}_b^y\right)|\bar{\psi}\rangle$$

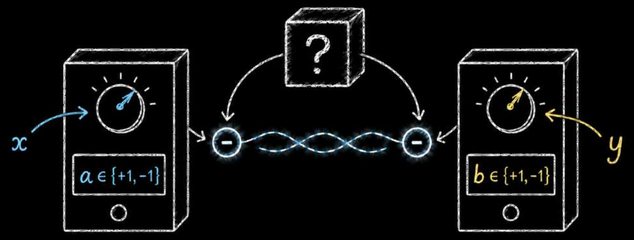
The state is unique !

$$\langle\psi|\Pi\left(M_a^x \otimes N_b^y\right)|\psi\rangle = \langle\bar{\psi}|\Pi\left(\bar{M}_a^x \otimes \bar{N}_b^y\right)|\bar{\psi}\rangle$$

All moments for two strategies are equal!

ABSTRACT SELF-TESTING

Paddock, Slofstra, Zhao, Zhou
AHP 25(10), [2024]



$$\{\mathfrak{m}_a^x, \mathfrak{n}_b^y\}$$

$$\phi$$

GNS representation: $\mathscr{H}_\phi, \pi_\phi, |\Omega_\phi\rangle$

Self-tested strategy: $\{\pi_\phi(\mathfrak{m}_a^x), \pi_\phi(\mathfrak{n}_b^y), |\Omega_\phi\rangle\}$

ALL MINIMAL GNS REPRESENTATIONS ARE UNITARILY EQUIVALENT. "JUNK" AND ISOMETRY COME FROM NON-MINIMAL GNS REPRESENTATIONS.

ABSTRACT SELF-TESTING : CONSEQUENCES

Algebraic definition can be adapted to different quantum models, for example commuting operator model, as opposed to the tensor product model

It allows to move more freely between different strategies, for example it identifies cases in which PVM assumption can be harmlessly introduced.

Algebraic definition keeps the essential algebraic content of strategies, it is essential for defining self-testing in other scenarios, for example sequential ones!

Robust self-testing defined as uniqueness of amenable tracial state!

Paddock, Slofstra, Zhao, Zhou
AHP 25(10), [2024]

Zhao
arXiv: 2411.03259, [2024]

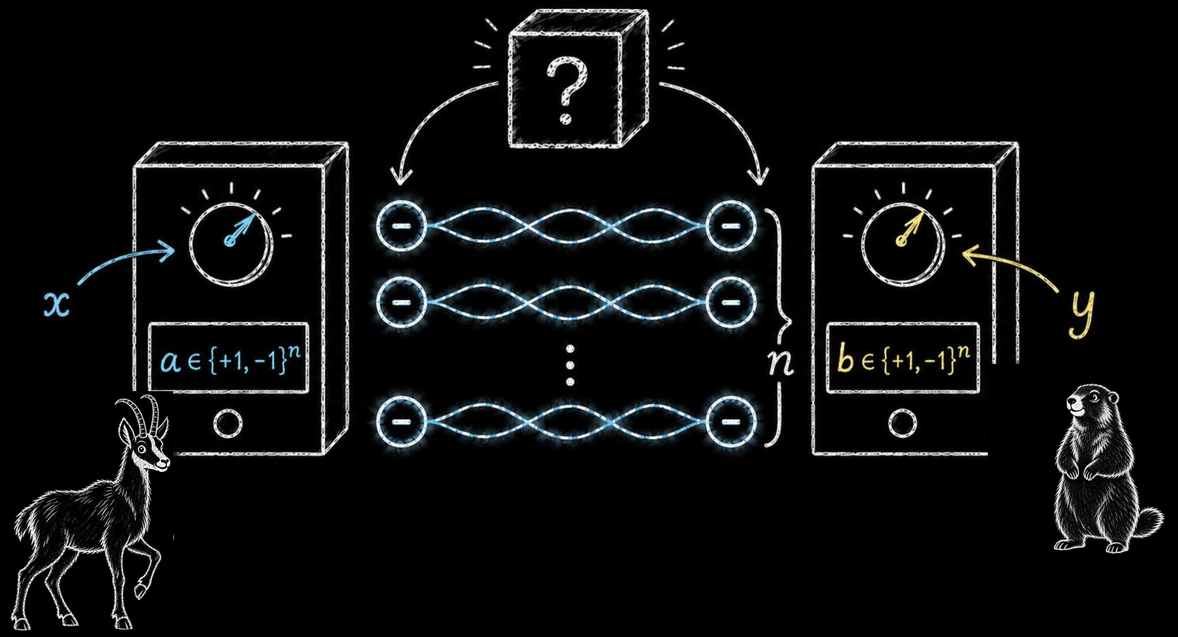
PARALLEL SELF-TESTING

Coladangelo
QIC 17 (9&10) [2017]

McKague
NJP 18 (4) [2016]

Šupić, Bowles, Cavalcanti
Quantum 5 (418) [2021]

Aim: certify that Alice and Bob share n EPR pairs



Approach: make them play n CHSH games, maximal score for all games robustly self-tests n EPR pairs

Problem: robustness function $f(\epsilon)$ depends also on n. Robustness decreases significantly when number of tested pairs increases.

EPR pairs are extracted one by one, hence robustness deteriorates with every new extracted pair.

$$\mathscr{H} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2$$

PARALLEL SELF-TESTING

Natarajan, Vidick
STOC [2017]

Instead of extracting pair by pair certify global algebraic relations

For all n-bit strings a and b on Alice's side certify existence of observables

$X(a), Z(b)$, which are in ideal case $\bigotimes_{i=1}^{n} \sigma_x^{a_i}$ and $\bigotimes_{i=1}^{n} \sigma_z^{b_i}$

These global observables satisfy the following reactions

$$X(a)X(b) = X(a+b) \qquad Z(a)Z(b) = Z(a+b) \qquad X(a)Z(b) = (-1)^{ab} Z(b)X(a)$$

Linearity in X | Linearity in Z | Anti-commutation

THESE RELATIONS ARE PRESENTATION OF n-QUBIT PAULI GROUP

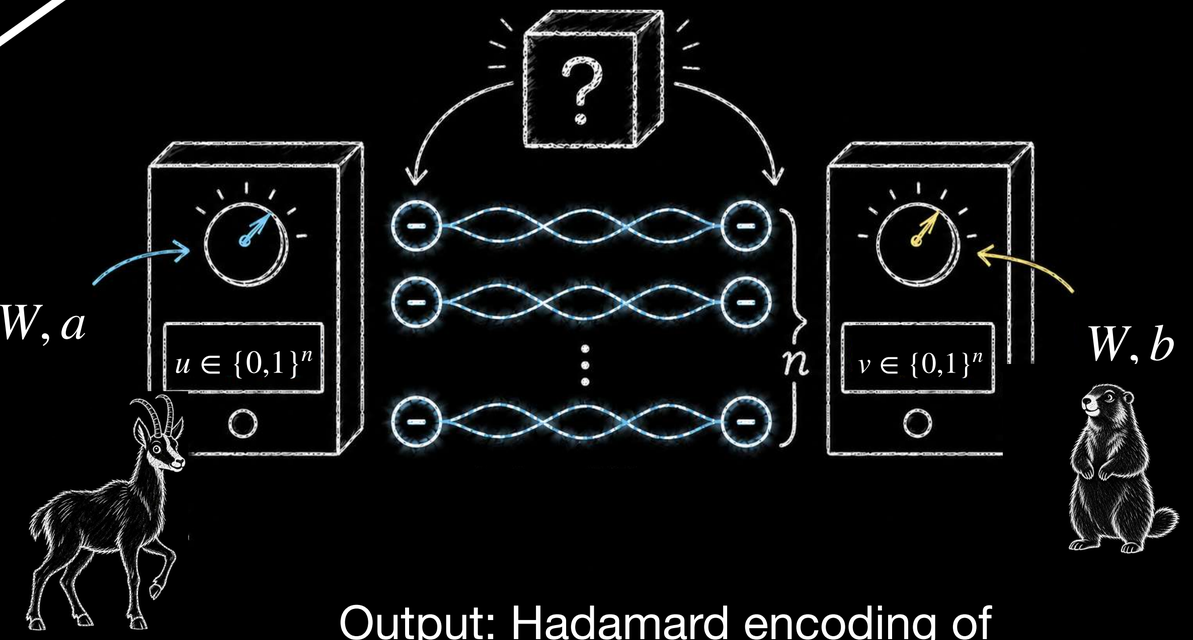
PARALLEL SELF-TESTING

Natarajan, Vidick
STOC [2017]

$$X(a)X(b) = X(a+b) \qquad Z(a)Z(b) = Z(a+b) \qquad X(a)Z(b) = (-1)^{\langle a,b \rangle} Z(b)X(a)$$

LINEARITY IN X | LINEARITY IN Z | ANTI-COMMUTATION

ORACULARIZED VERSION OF BLR LINEARITY TESTING



MAGIC SQUARE GAME

Input: Basis type W and n-bit string a/b

Input length: $2n$

Output: Hadamard encoding of the obtained outputs u and v: $\alpha = (-1)^{a \cdot u}$ and $\beta = (-1)^{b \cdot v}$

Output length: 2

GH gives robustness that does not depend on n because we work with global observables

PARALLEL SELF-TESTING

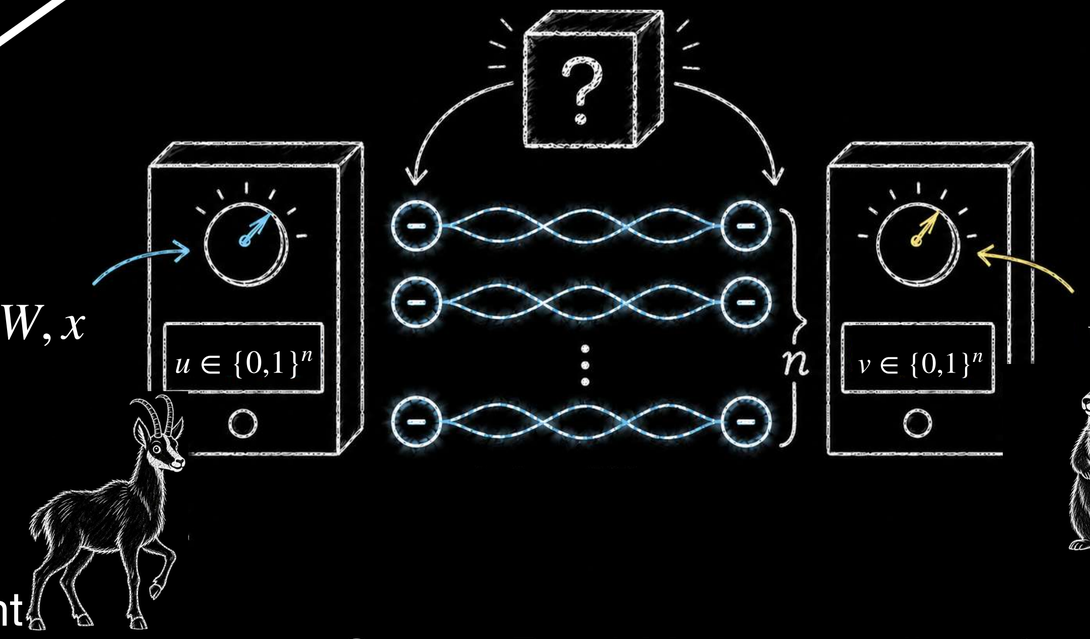
Natarajan, Vidick
FOCS [2018]

$$X(a)X(b) = X(a+b) \qquad Z(a)Z(b) = Z(a+b) \qquad X(a)Z(b) = (-1)^{\langle a,b \rangle} Z(b)X(a)$$

LINEARITY IN X | LINEARITY IN Z | ANTI-COMMUTATION

$a, b \in \mathbb{F}_q^N$

RAZ-SAFRA PLANE-VS-POINT LOW-DEGREE TESTING



MAGIC SQUARE GAME

Input: Basis type W and a point
$x \in \mathbb{F}_q^N$ for one player or a plane
P from $\mathbb{F}_q^N$ for the other
Input length: $O(\log n)$

Output: Reed-Muller encoding of obtained outputs u and v

Output length: $poly(\log \log n)$

GH gives robustness that does not depend on n because we work with global observables

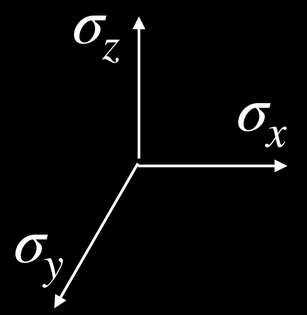
★ ACT III : SOME APPLICATIONS

APPLICATIONS IN CRYPTOGRAPHY

PRIVATE DI RANDOMNESS AND QKD

Acin, Pironio, Vertesi, Wittek
PRA 93 [2016]

- MAXIMAL RANDOMNESS FROM QUBIT STATES
- GISIN'S ELEGANT BELL INEQUALITY SELF-TESTS THREE PAULIS ON ALICE'S SIDE AND AN EPR PAIR
- THREE PAULIS ARE TOMOGRAPHICALLY COMPLETE





$|\mathrm{EPR}\rangle$

$$\langle \mathrm{EPR} | A_x \otimes B_y | \mathrm{EPR} \rangle = \frac{1}{2} \mathrm{Tr}[A_x B_y^T]$$

B_y IS A FOUR-OUTCOME POVM (TETRAHEDRON)

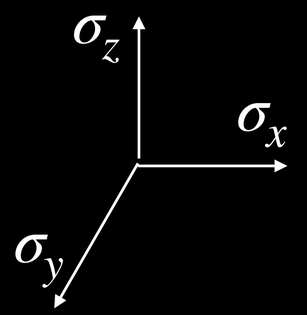
APPLICATIONS IN CRYPTOGRAPHY

PRIVATE DI RANDOMNESS AND QKD

Acin, Pironio, Vertesi, Wittek
PRA 93 [2016]

Farkas, Volcic, Storgaard, Chen, Mancinska
Nat. Phys. 22, 2 [2026]

- MAXIMAL RANDOMNESS FROM QUBIT STATES
- GISIN'S ELEGANT BELL INEQUALITY SELF-TESTS THREE PAULIS ON ALICE'S SIDE AND AN EPR PAIR
- THREE PAULIS ARE TOMOGRAPHICALLY COMPLETE





$|\mathrm{EPR}\rangle$

$2\log(d)$

$$\langle \mathrm{EPR}|A_x \otimes B_y|\mathrm{EPR}\rangle = \frac{1}{2}\mathrm{Tr}[A_x B_y^T]$$

B_y IS A FOUR-OUTCOME POVM (TETRAHEDRON)

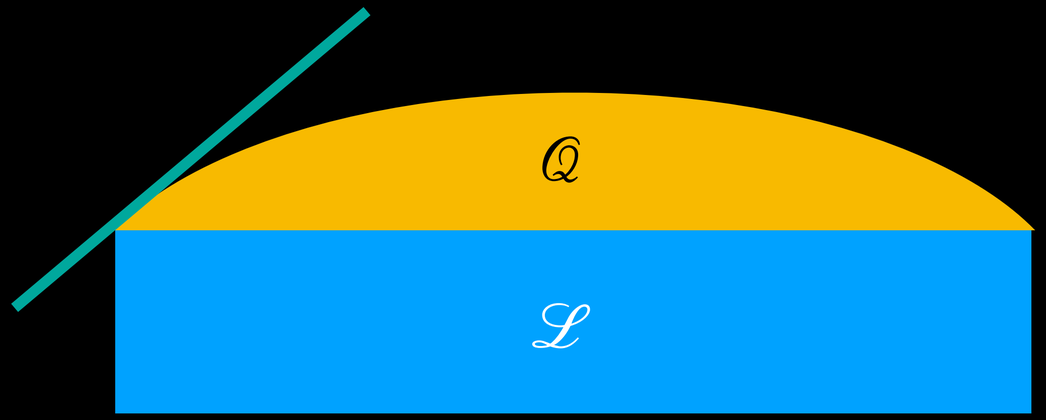
APPLICATIONS IN CRYPTOGRAPHY

PRIVATE RANDOMNESS AND QKD

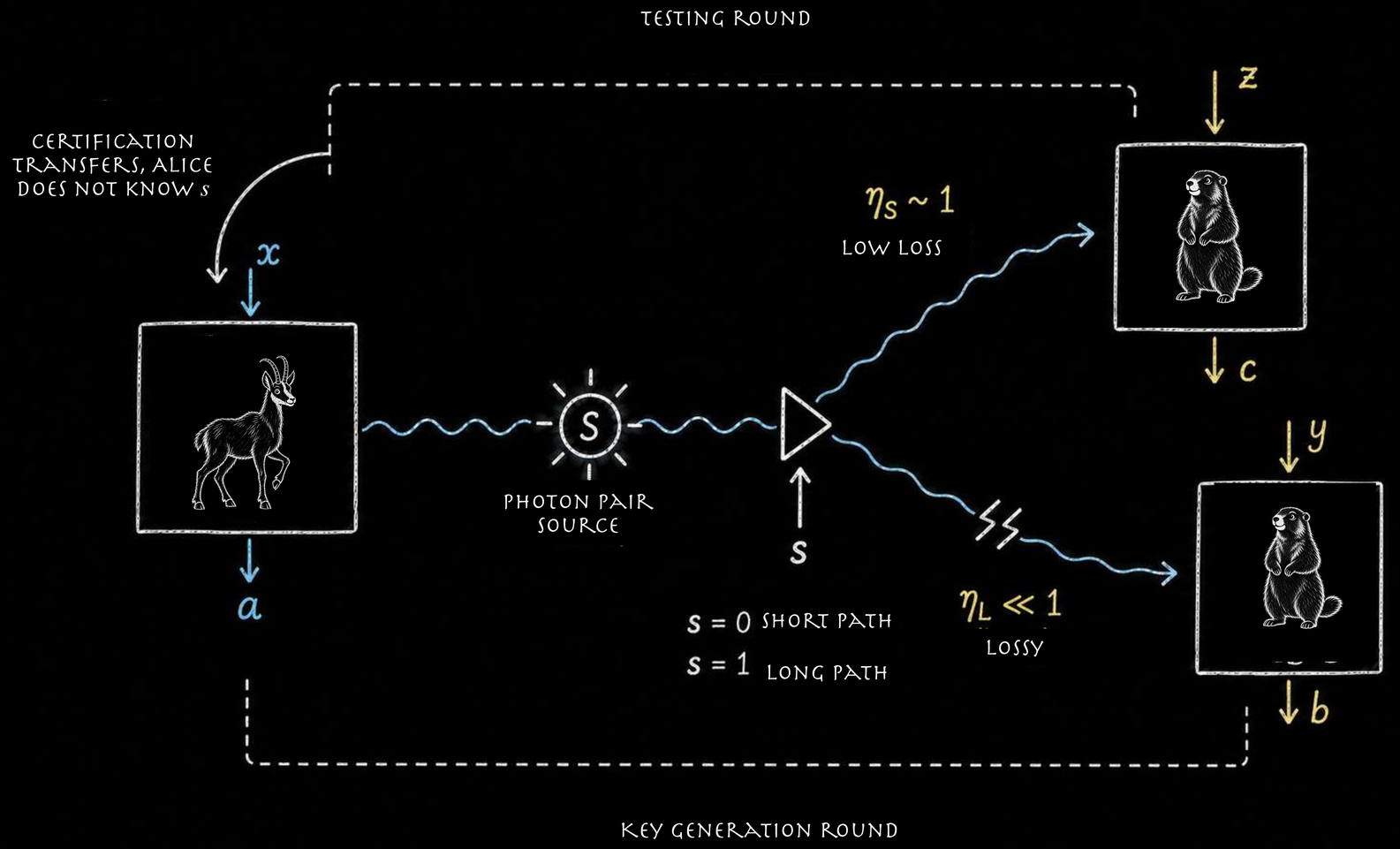
- MAXIMAL RANDOMNESS FROM ARBITRARY SMALL NONLOCALITY
- BELL OPERATOR: $\cos(\theta+\phi)\cos(\theta+\omega)A_0(\cos\omega B_0-\cos\phi B_1)+\cos\phi\cos\omega A_1\left(-\cos(\theta+\omega)B_0+\cos(\theta+\phi)B_1\right)$
- MAXIMAL VIOLATION SELF-TESTS THE STRATEGY: $|\psi\rangle=\frac{|00\rangle+i|11\rangle}{\sqrt{2}}$, $A_0=\sigma_x$, $A_1=\cos\theta\sigma_x+\sin\theta\sigma_y$

  $B_0=\cos\phi\sigma_x+\sin\phi\sigma_y$, $B_1\cos\omega\sigma_x+\sin\omega\sigma_y$

Wooltorton, Brown, Colbeck
PRL 132(21) [2024]



APPLICATIONS IN CRYPTOGRAPHY



1. SHORT RANGE TEST REACHES A HIGH CHSH VALUE: SELF-TEST OF THE SOURCE AND ALICE'S MEASUREMENTS

2. THE SOURCE AND ALICE'S DEVICES ARE THE SAME IN BOTH CONFIGURATIONS

3. EFFECTIVELY KEY IS GENERATED IN A SEMI-DEVICE-INDEPENDENT MANNER

Chaturvedi, Viola, Pawlowski
npj QI 10, 7 [2024]

Lobo, Pauwels, Pironio
Quantum 8, 1332 [2024]

Le Roy-Deloison, Lobo, Pauwels, Pironio
PRX Quantum 6, 020311 [2025]

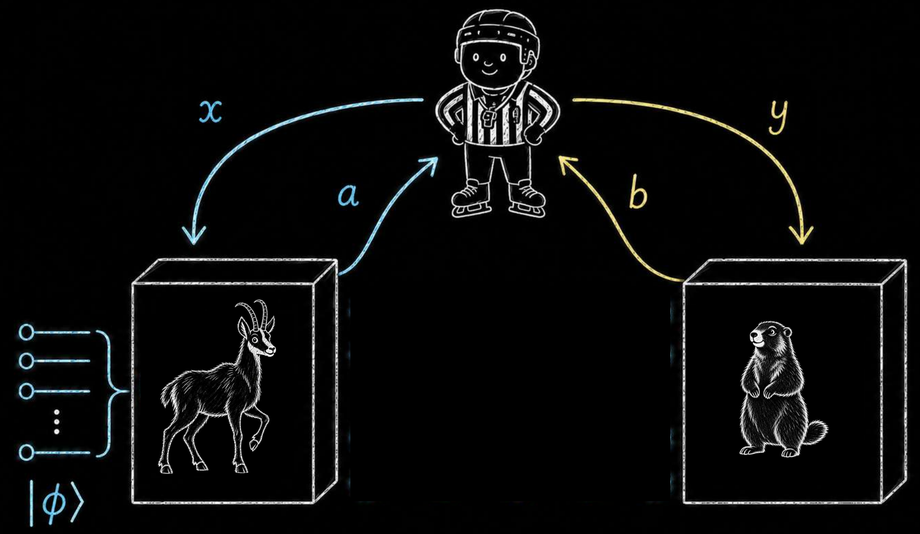
PROOF OF QUANTUM KNOWLEDGE

A PROOF OF KNOWLEDGE CONVINCES A VERIFIER THAT THE PROVER HAS SOMETHING, A PRIVATE KEY OR A CREDENTIAL, WITHOUT REVEALING IT.

FORMALISED BY AN EXTRACTOR: GIVEN ORACLE ACCESS TO ANY PROVER THAT MAKES THE VERIFIER ACCEPT IT RECONSTRUCTS THE SECRET.

QUANTUM VERSION:

1. WITNESS IS A QUANTUM STATE (GROUND STATE OF A LOCAL HAMILTONIAN)
2. THE VERIFIER IS CLASSICAL, IT CANNOT RECEIVE A QUBIT
3. THE WITNESS CANNOT BE COPIES

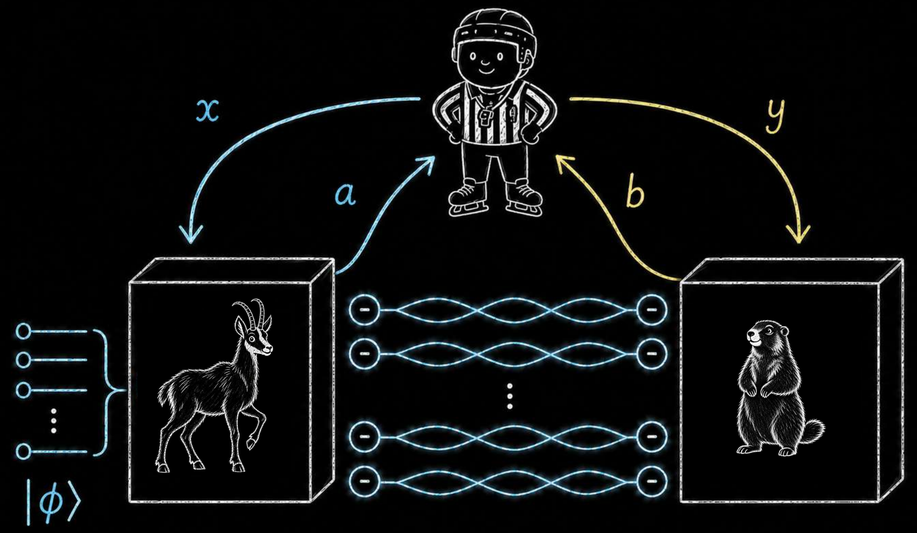


PROOF OF QUANTUM KNOWLEDGE

SELF-TESTING ROUND: VERIFIER ENSURES THAT PROVERS SHARE n EPR PAIRS AND PAULI MEASUREMENTS

ENERGY ROUND: PROVER ALICE IS ASKED TO TELEPORT HER QUANTUM WITNESS TO BOB, AND BOB GETS THE SAME QUESTION AS IN THE SELF-TESTING ROUND.

EXTRACTION ISOMETRY (A PARTICULAR ONE, THE SWAP GATE) IS USED AS THE EXTRACTOR OF THE WITNESS!



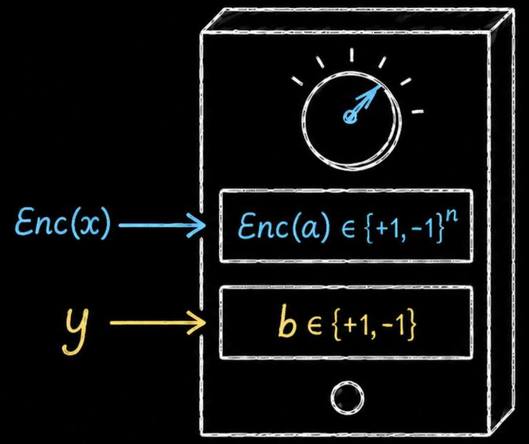
Broadbent, Grilo, Hara,Mehta,
arXiv: 2503.13699 [2025]

COMPILED NONLOCAL GAMES

PROOFS OF QUANTUMNESS AND A WAY OUT OF THE TWO-DEVICE PROBLEM

NONLOCAL GAMES REQUIRE SPATIAL SEPARATION, A CONDITION THAT IS NOT EASY TO IMPOSE, AND IN MANY ARCHITECTURES IT IS NOT POSSIBLE.

Kalai, Lombardi, Vaikuntanathan, Yang
STOC [2023]



IND-CPA SECURITY OF QHE ENCRYPTION ENSURES THAT THE PROVER CANNOT GUESS x BETTER THAN RANDOMLY GUESSING. IN FACT THE SECURITY EMULATES THE LOCALITY CONDITION FROM BELL TESTS!

FOR ALL GAMES CLASSICAL BOUND IS PRESERVED THROUGH THE COMPILATION, AND FOR MOST GAMES (NON-PATHOLOGICAL ONES) THE QUANTUM BOUND IS PRESERVES ALSO.

RIGIDITY OF COMPILED NONLOCAL GAMES

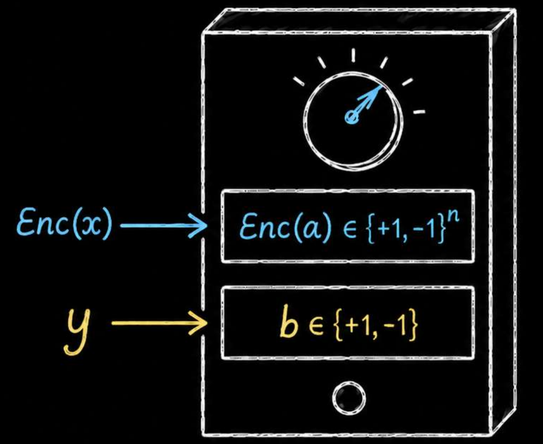
PROOFS OF QUANTUMNESS AND A WAY OUT OF THE TWO-DEVICE PROBLEM

WHY IS IT HARD?

NO TENSOR PRODUCT STRUCTURE

HOW TO DEFINE LOCALITY AND ENTANGLEMENT FOR A SINGLE DEVICE?

HOW TO DEAL WITH ENCRYPTED OPERATIONS?



Natarajan, Zhang
FOCS [2023]

Cui, Malavolta, Mehta, Natarajan, Paddock, Schmidt, Walter, Zhang
Quantum 10 [2026]

Baroni, Vu, Bourdoncle, Diamanti, Markham, Šupić
Quantum 9 [2025]

Mehta, Paddock, Wooltorton
TQC [2025]

WHAT IS KNOWN?

THE STATE COMING TO THE SECOND ROUND AND THE MEASUREMENTS PERFORMED IN THE SECOND ROUND CAN BE SELF-TESTED FOR A FAIRLY LARGE CLASS OF GAMES

SUCCINT ARGUMENT FOR QMA

COMMUNICATION AND VERIFIER'S TIME ARE *polylog* IN THE SIZE OF COMPUTATION

SOUND AGAINST COMPUTATIONALLY BOUNDED PROVERS

QUANTUM NPA

CLAIM: THIS HAMILTONIAN HAS A LOW-ENERGY GROUND STATE
WITNESS: QUANTUM STATE

Metger, Natarajan, Zhang
FOCS [2024]

SUCCINT ARGUMENT FOR QMA

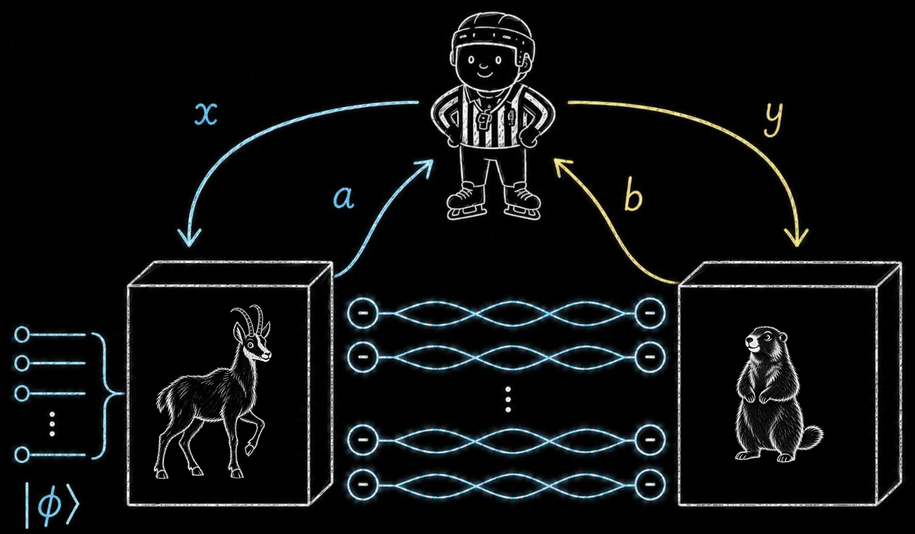
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Metger, Natarajan, Zhang
FOCS [2024]

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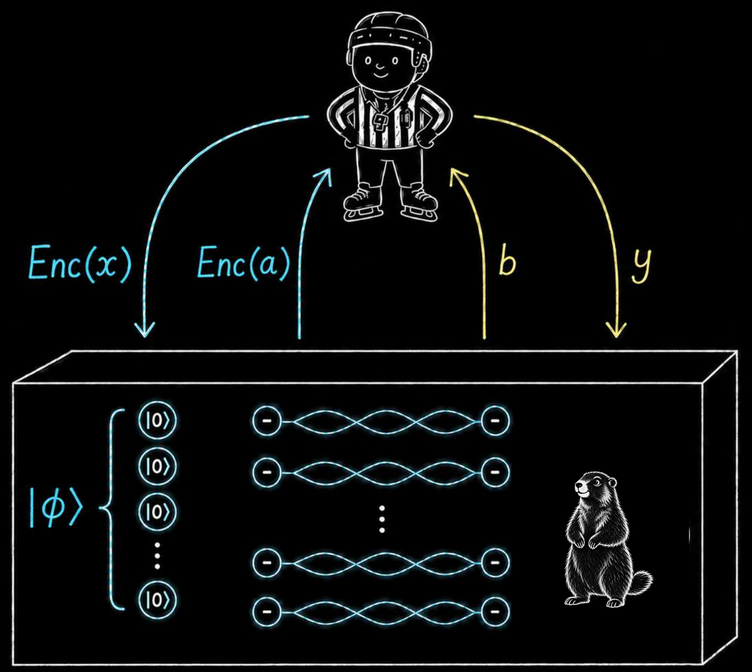
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Metger, Natarajan, Zhang
FOCS [2024]

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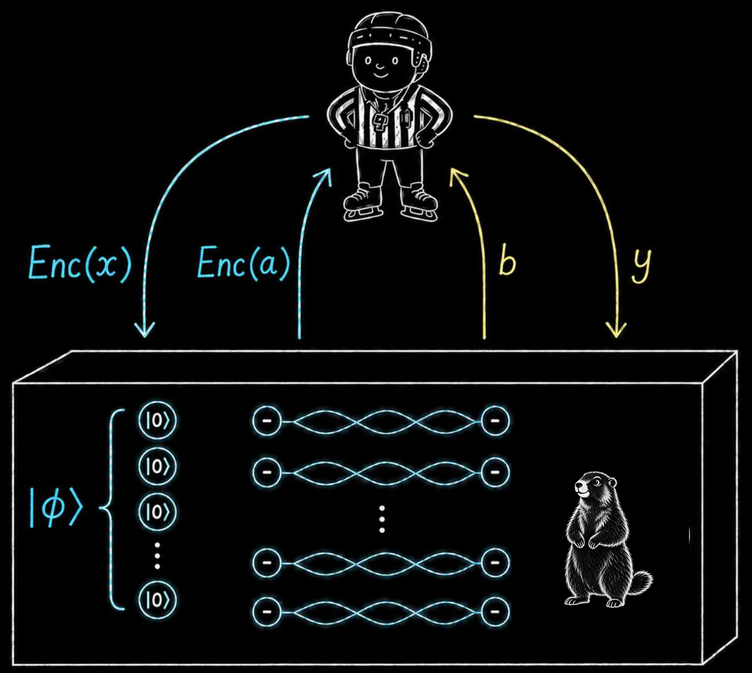
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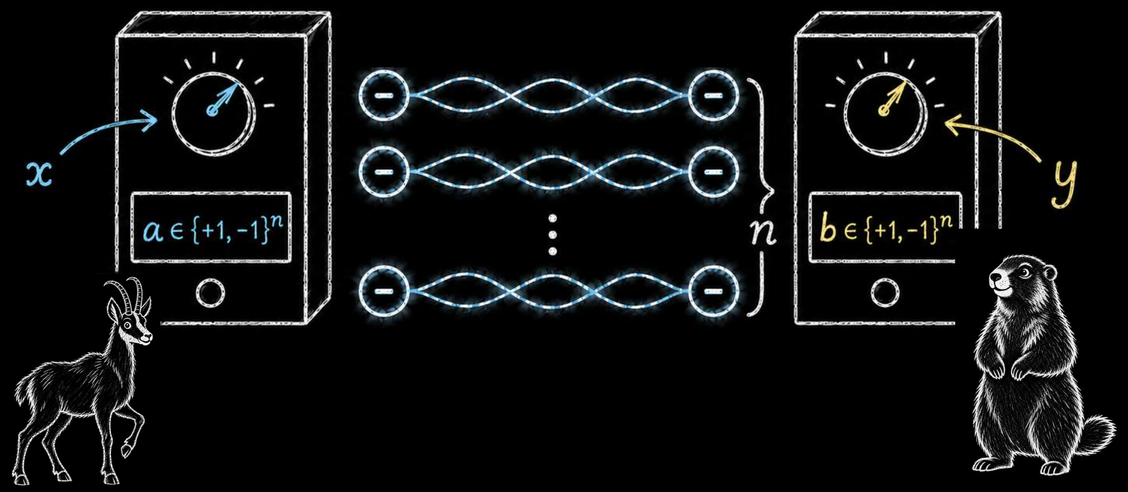


THE PROTOCOL IS SIMILAR IN STRUCTURE TO THE ONE PROVIDING THE PROOF OF KNOWLEDGE.

THE GAMES USED FOR SELF-TESTING ARE COMPILED, AND RIGIDITY OF THE SECOND ROUND OPERATIONS IS ENOUGH FOR THE CONSTRUCTION OF THE ARGUMENT!

Metger, Natarajan, Zhang
FOCS [2024]

VERIFICATION OF COMPUTING



Reichardt, Unger, Vazirani
Nature 496(7446) [2013]

Coladangelo, Grilo, Jeffery, Vidick
TACT [2019]

Broadbent, Mehta, Zhao
arXiv: 2304.03448 [2023]

TESTING ROUND

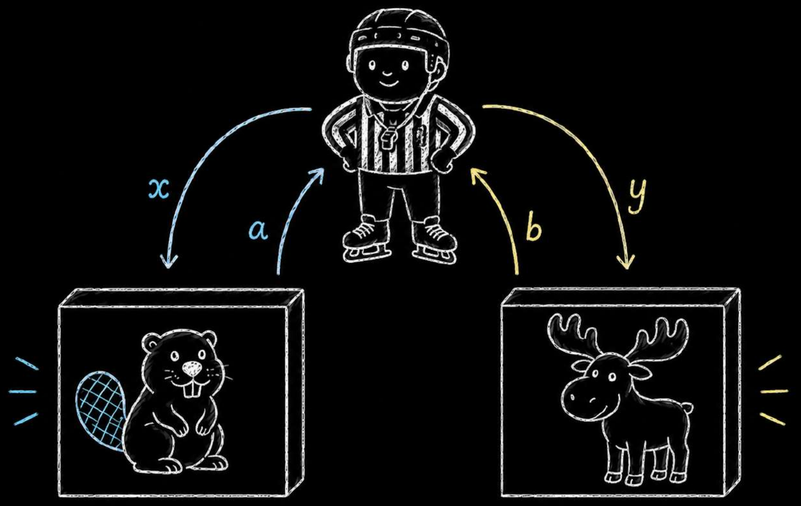
SELF-TEST THE STATE AND MEASUREMENTS THAT ALLOW UNIVERSAL QUANTUM COMPUTING

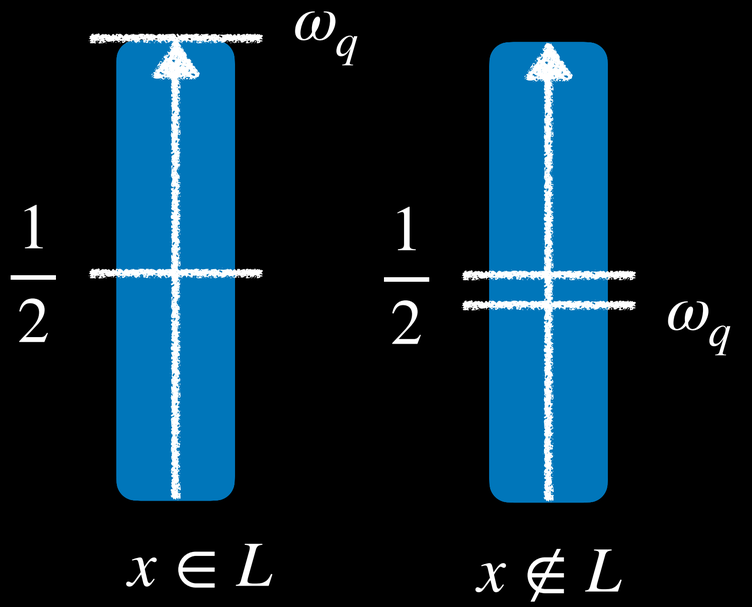
COMPUTATION ROUND

GIVE THE PROVERS THE INPUT FOR COMPUTATION

ALL THE MEASUREMENTS USED IN COMPUTATION HAVE TO BE SELF-TESTED !

APPLICATIONS IN COMPLEXITY

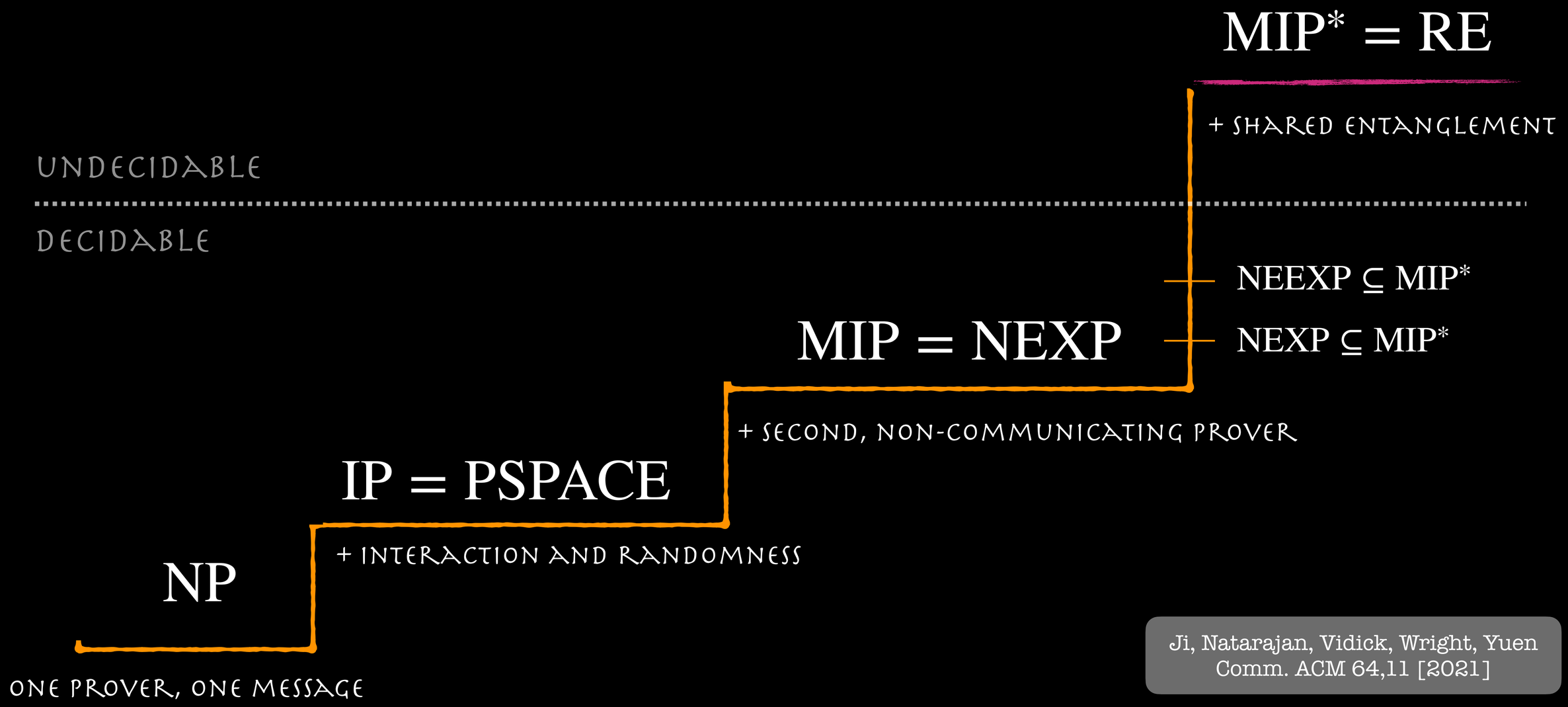




$L \in \mathrm{MIP}^*$ IFF THERE IS A POLY-TIME MAP $x \to G_x$ WITH $\omega_q(G_x) = 1$ WHEN $x \in L$, AND $\omega_q(G_x) \leq \frac{1}{2}$ WHEN $x \notin L$.

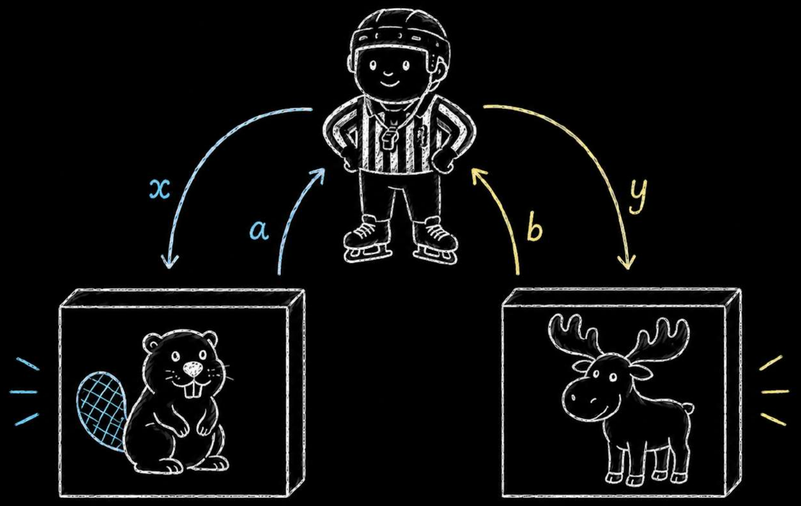
GIVEN ANY TURING MACHINE M, YOU CAN EFFICIENTLY WRITE DOWN A NONLOCAL GAME THAT HAS A PERFECT QUANTUM STRATEGY EXACTLY WHEN M HALTS.

APPLICATIONS IN COMPLEXITY



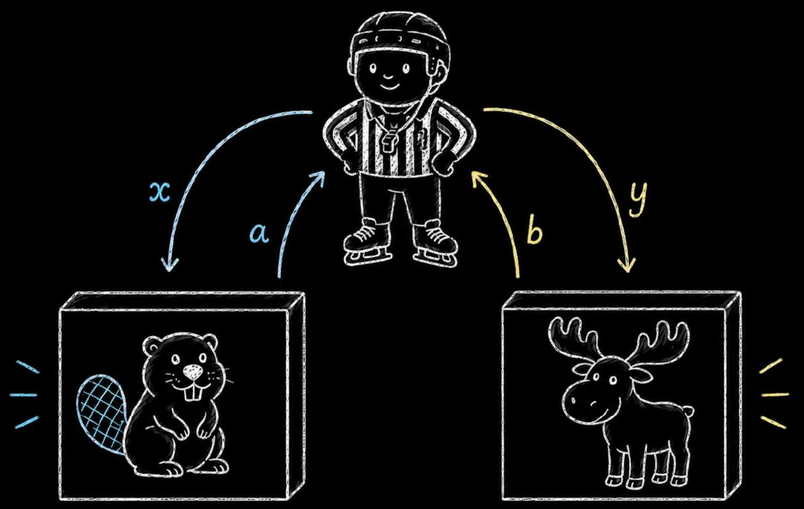
Ji, Natarajan, Vidick, Wright, Yuen
Comm. ACM 64,11 [2021]

APPLICATIONS IN COMPLEXITY



PROVERS HAVE ADDITIONAL RESOURCE: ENTANGLEMENT WHICH COULD HELP THEM TO CHEAT, SOUNDNESS SHOULD BREAK?

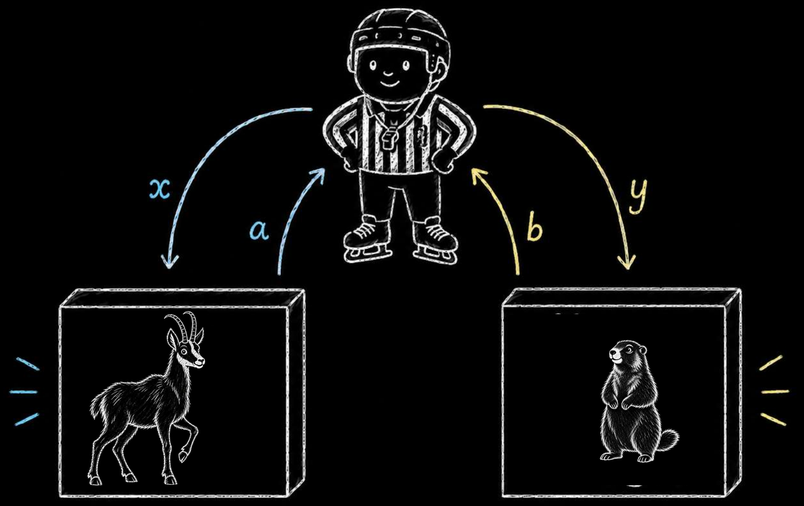
APPLICATIONS IN COMPLEXITY



1. FORCE PROVERS TO USE ENTANGLEMENT IN A CERTAIN WAY THAT WILL NOT ALLOW THEM TO CHEAT!

PROVERS HAVE ADDITIONAL RESOURCE: ENTANGLEMENT WHICH COULD HELP THEM TO CHEAT, SOUNDNESS SHOULD BREAK?

APPLICATIONS IN COMPLEXITY

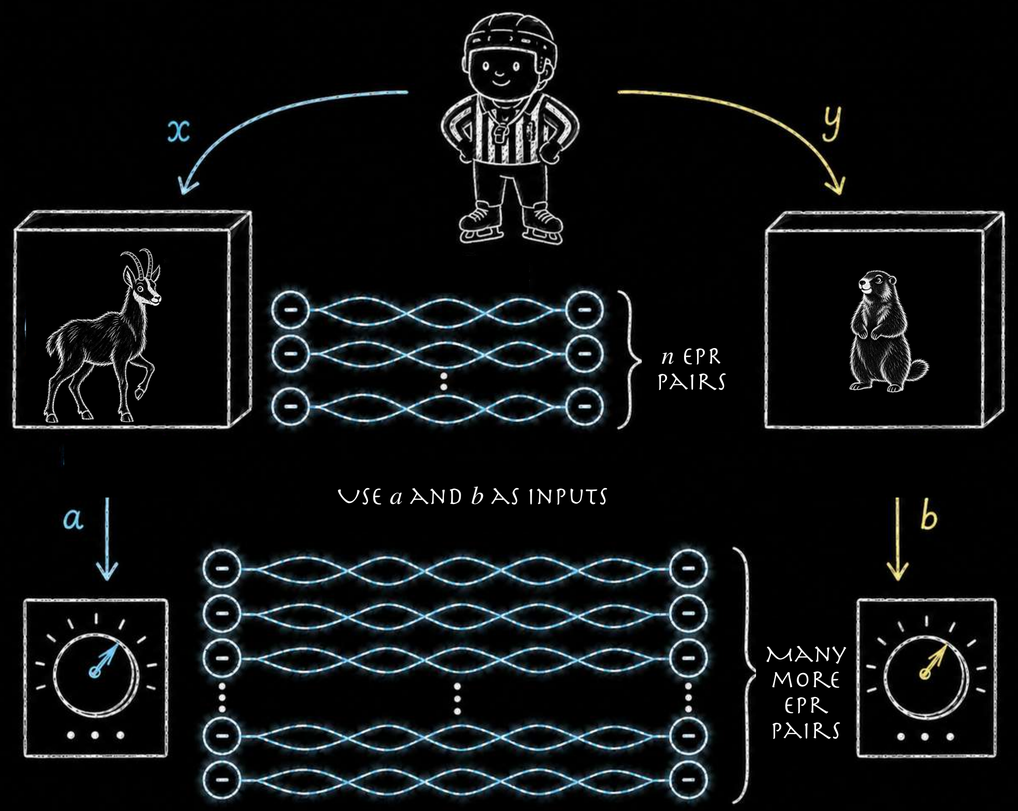


PROVERS HAVE ADDITIONAL RESOURCE: ENTANGLEMENT WHICH COULD HELP THEM TO CHEAT, SOUNDNESS SHOULD BREAK?

1. FORCE PROVERS TO USE ENTANGLEMENT IN A CERTAIN WAY THAT WILL NOT ALLOW THEM TO CHEAT!

2. INPUTS FOR PROBLEMS WHOSE SOLUTIONS VERIFIER NEEDS TO VERIFY ARE TOO LONG, EVEN TO JUST BE READ, PARALLEL SELF-TESTING PROVIDES SOLUTION: EXPONENTIALLY LARGE SYSTEM CAN BE VERIFIED WITH "SHORT INPUTS"

APPLICATIONS IN COMPLEXITY



INTROSPECTION

PROVERS ARE DOING THE VERIFIER'S JOB, THEY ARE SAMPLING A CORRECT INPUT DISTRIBUTION BY THEMSELVES!

THEY DO NOT KNOW EACH OTHER'S INPUTS BECAUSE THEY MEASURE IN THE COMPLEMENTARY BASES.

WHAT CAN WE HEAR?

THE QUESTION WE STARTED WITH

CAN QUANTUM THEORY PASS THE TEST OF CARTESIAN METHODOLOGICAL DOUBT?

YES - UP TO LOCAL ISOMETRY!

THAT RESIDUAL FREEDOM IS NOT A WEAKNESS, BUT A DOUBT NO EXPERIMENT CAN EVER REMOVE.

STILL OPEN

- ROBUSTNESS THAT DOES NOT DECAY WITH THE SYSTEM SIZE
- FIDELITY ACCUMULATION THEOREM
- SELF-TESTING OF MULTIPARTITE STATES WITH CONSTANT NUMBER OF INPUTS
- SELF-TESTING OF MULTIPARTITE QUDIT STATES
- MORE TECHNIQUES FOR ROBUST SELF-TESTING

SELF-TESTING BEGAN AS A QUESTION OF TRUSTING A DEVICE
IT ENDED-UP DESCRIBING THE LIMITS OF WHAT CAN BE COMPUTED